FACULTY OF ENGINEERING • KHON KAEN UNIVERSITY

Fundamental Electrical Engineering (FunE)

Final Exam Master Flashcards & Solved Problems Booklet (78 Cards)

COURSE & INSTRUCTOR
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COVERAGE
Chapters 6, 7, 8, 9 & 10 (100% Complete Syllabus)

CONTENT INCLUSIONS
78 Cards • Authentic Slide & Exam Diagrams • Line-by-Line Math

PART 1

Chapter 6: Three-Phase Circuits

Balanced & Unbalanced Y-Y, Y-Δ, Complex Power, Neutral Shift, 2-Wattmeter Method (10 Flashcards)

CH06-01 | Chapter 6 | Level 1 (Core Concept) | 📌 Lecture Slides & Formula Sheet

Star (Y) vs Delta (Δ) Voltage & Current Relations

📌 **PROBLEM / QUESTION:**

In a balanced 3-phase positive-sequence (abc) system, state the exact relationships between line quantities (V_L, I_L) and phase quantities (V_p, I_p) for both Y-connection and Δ-connection.

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Star (Y)}: V_L = \sqrt{3}\, V_p \angle +30^\circ, \quad I_L = I_p$$
$$\text{Delta (Δ)}: V_L = V_p, \quad I_L = \sqrt{3}\, I_p \angle -30^\circ$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Star (Y) Connection:
- Line Voltage leads Phase Voltage by 30°:

$$V_{ab} = \sqrt{3}\, V_{an} \angle +30^\circ$$

- Line Current equals Phase Current:

$$I_L = I_p$$

2. Delta (Δ) Connection:
- Line Voltage equals Phase Voltage:

$$V_L = V_p$$

- Line Current lags Phase Current by 30°:

$$I_a = \sqrt{3}\, I_{ab} \angle -30^\circ$$

⚠️ **EXAM PITFALL & CAUTION:**

In negative sequence (acb), line voltage leads by -30° (or lags by 30°). Remember: in Y, currents are identical; in Δ, voltages are identical.

CH06-02 | Chapter 6 | Level 1 (Core Concept) | 📌 HW Ch.6 Problem 1 & 5

Delta to Wye (Δ ↔ Y) Impedance Conversion

📌 **PROBLEM / QUESTION:**

If a balanced 3ϕ load is connected in Δ with impedance $Z_\Delta = 60 + j30\ \Omega$, what is the equivalent impedance per phase when converted to a Wye (Y) connection (Z_Y)?

📐 **KEY FORMULAS & DEFINITIONS:**

$$Z_Y = \frac{Z_\Delta}{3} \quad \Leftrightarrow \quad Z_\Delta = 3Z_Y$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

$$Z_Y = \frac{Z_\Delta}{3} = \frac{60 + j30}{3} = 20 + j10\ \Omega$$

Conversely, converting from Wye to Delta: $Z_\Delta = 3Z_Y$.

⚠️ **EXAM PITFALL & CAUTION:**

Multiplying by 3 instead of dividing is a classic mistake! Delta load draws 3 times more current than Wye load of the same impedance, so equivalent Wye impedance must be 3 times smaller.

CH06-03 | Chapter 6 | Level 1 (Core Concept) | 📌 Formula Sheet & Lecture Slides

Three-Phase Power Formulas & Complex Power

📌 **PROBLEM / QUESTION:**

State the formulas for balanced 3ϕ Real Power ($P_{3\phi}$), Reactive Power ($Q_{3\phi}$), Apparent Power ($|S_{3\phi}|$), and Complex Power ($S_{3\phi}$) in terms of line values (V_L, I_L).

📐 **KEY FORMULAS & DEFINITIONS:**

$$P_{3\phi} = \sqrt{3}V_L I_L \cos\theta \quad [\text{W}] \quad Q_{3\phi} = \sqrt{3}V_L I_L \sin\theta \quad [\text{VAR}] \quad |S_{3\phi}| = \sqrt{3}V_L I_L \quad [\text{VA}] \quad S_{3\phi} = \sqrt{3}V_L I_L^* = P + jQ$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

When using Line values (V_L, I_L), always include the factor $\sqrt{3}$.
When using Phase values (V_p, I_p), use factor 3:
- $P = 3V_p I_p \cos\theta$
- $Q = 3V_p I_p \sin\theta$
- $S = 3V_p I_p^*$

where $\theta = \angle Z_p$ is the load phase impedance angle, and $PF = \cos\theta$.

⚠️ **EXAM PITFALL & CAUTION:**

Never mix $\sqrt{3}$ with phase quantities! It is either $\sqrt{3}V_L I_L$ or $3V_p I_p$.

CH06-04 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 1 (Page 1)

Balanced 4-Wire Y-Y System Full Calculation

PROBLEM / QUESTION:

In the balanced Y-Y circuit shown below, source phase voltages are $V_{an} = 220\angle 0^\circ$ V, $V_{bn} = 220\angle -120^\circ$ V, $V_{cn} = 220\angle -240^\circ$ V. Line impedance $Z_{\text{line}} = 10 + j5\ \Omega$. Balanced load impedance $Z_L = 20 + j50\ \Omega$. Find: (1) Line voltages V_{ab}, V_{bc}, V_{ca}, (2) Line currents I_{aA}, I_{bB}, I_{cC}, (3) Neutral current I_{Nn}, (4) Total 3ϕ complex power S.

active power and power factor.

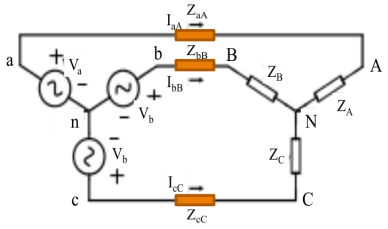


Fig. 1 Y to Y circuit with line impedance and balanced load.

รูปประกอบ: Homework 6 Figure 1: Balanced Y-Y System

KEY FORMULAS & DEFINITIONS:

$V_{ab} = \sqrt{3}V_{an}\angle +30^\circ$ $I_{aA} = \frac{V_{an}}{Z_{\text{line}}+Z_L}$ $I_{Nn} = 0$ $S_{3\phi} = 3V_{an}I_{aA}^*$

STEP-BY-STEP MODEL SOLUTION:

1. Line Voltages:
- $V_{ab} = \sqrt{3}(220)\angle(0^\circ + 30^\circ) = 381.05\angle 30^\circ$ V
- $V_{bc} = 381.05\angle -90^\circ$ V
- $V_{ca} = 381.05\angle -210^\circ$ V

2. Per-phase total impedance:

$$Z_{\text{total}} = (10 + j5) + (20 + j50) = 30 + j55\ \Omega = 62.65\angle 61.39^\circ\ \Omega$$

3. Line Currents:
- $I_{aA} = \frac{220\angle 0^\circ}{62.65\angle 61.39^\circ} = 3.51\angle -61.39^\circ$ A
- $I_{bB} = 3.51\angle -181.39^\circ$ A
- $I_{cC} = 3.51\angle 58.61^\circ$ A

4. Neutral Current:
In a balanced system, $I_{Nn} = 0$ A, $V_{Nn} = 0$ V.

5. Total 3ϕ Power:

$$S = 3 \times 220 \times 3.51\angle +61.39^\circ = 2,316.6\angle 61.39^\circ\ \text{VA} = 1,109\ \text{W} + j2,034\ \text{VAR}$$

EXAM PITFALL & CAUTION:

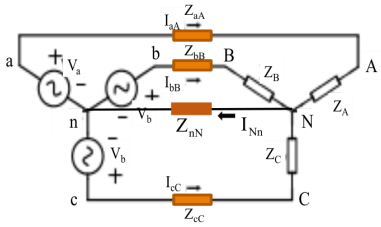
Always remember to conjugate the current (I^*) when calculating complex power: $S = VI^*$. The angle of S equals the angle of Z.

CH06-05 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 2 (Page 2)

Unbalanced 4-Wire Y-Y System with Neutral Wire

PROBLEM / QUESTION:

In the 4-wire Y-Y system shown below, the source is balanced $220\ \text{V}$ rms, line impedance $Z_{\text{line}} = 10 + j5\ \Omega$, neutral line impedance $Z_{nN} = 10 + j5\ \Omega$, and the load is UNBALANCED: $Z_A = 20 + j50\ \Omega$, $Z_B = 10 + j50\ \Omega$, $Z_C = 20 + j20\ \Omega$. How do you solve for the Neutral Shift Voltage (V_{Nn}) and line currents?



ig. 2 Y to Y circuit with line impedance and unbalanced load.

ed from: https://en.wikipedia.org/wiki/Three-phase_electric_power

รูปประกอบ: Homework 6 Figure 2: Unbalanced Y-Y 4-Wire with Neutral Line

KEY FORMULAS & DEFINITIONS:

$$V_{Nn} = \frac{\frac{V_{an}}{Z'_a} + \frac{V_{bn}}{Z'_b} + \frac{V_{cn}}{Z'_c}}{\frac{1}{Z'_a} + \frac{1}{Z'_b} + \frac{1}{Z'_c} + \frac{1}{Z_{nN}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Millman's / Node Voltage at Neutral node N:
Let node n be the reference ($0\ \text{V}$).
Total phase branch impedances:
- $Z'_a = (10 + j5) + (20 + j50) = 30 + j55\ \Omega$
- $Z'_b = (10 + j5) + (10 + j50) = 20 + j55\ \Omega$
- $Z'_c = (10 + j5) + (20 + j20) = 30 + j25\ \Omega$
- $Z_{nN} = 10 + j5\ \Omega$

2. Solve for V_{Nn} using KCL at node N:

$$\frac{V_N - V_{an}}{Z'_a} + \frac{V_N - V_{bn}}{Z'_b} + \frac{V_N - V_{cn}}{Z'_c} + \frac{V_N}{Z_{nN}} = 0$$

3. Calculate Line Currents:
- $I_{aA} = \frac{V_{an} - V_{Nn}}{Z'_a}$
- $I_{bB} = \frac{V_{bn} - V_{Nn}}{Z'_b}$
- $I_{cC} = \frac{V_{cn} - V_{Nn}}{Z'_c}$
- $I_{Nn} = \frac{V_{Nn}}{Z_{nN}} = I_{aA} + I_{bB} + I_{cC}$

EXAM PITFALL & CAUTION:

Do NOT assume $V_{Nn} = 0$! In an unbalanced system, current flows through the neutral wire, causing a non-zero neutral shift voltage $V_{Nn} \neq 0$.

CH06-06 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 3 (Page 2)

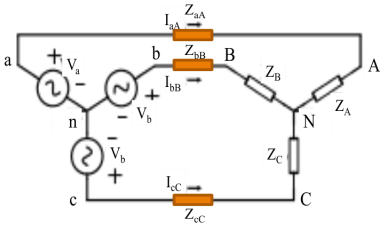
Unbalanced 3-Wire Y-Y System (NO Neutral Wire)

PROBLEM / QUESTION:

In the 3-wire Y-Y system shown below (NO neutral wire), the load is unbalanced ($Z_A \neq Z_B \neq Z_C$). How is the neutral shift voltage V_{Nn} calculated, and what happens to the neutral impedance term in the Millman formula?

ents (I_{aA}, I_{bB}, I_{cC}), I_{Nn}, phase voltages (V_{AN}, V_{BN}, V_{CN}) of loads, 3ϕ complex

al power, 3ϕ reactive power and power factor.



รูปประกอบ: Homework 6 Figure 3: Unbalanced Y-Y 3-Wire System

KEY FORMULAS & DEFINITIONS:

$$V_{Nn} = \frac{\frac{V_{an}}{Z'_a} + \frac{V_{bn}}{Z'_b} + \frac{V_{cn}}{Z'_c}}{\frac{1}{Z'_a} + \frac{1}{Z'_b} + \frac{1}{Z'_c}}$$

$$I_{aA} + I_{bB} + I_{cC} = 0$$

STEP-BY-STEP MODEL SOLUTION:

1. Floating Neutral Point:

Because there is no neutral conductor, $Z_{nN} = \infty$, meaning the admittance term $\frac{1}{Z_{nN}} = 0$.

2. Node Voltage equation at N:

$$V_{Nn} = \frac{V_{an}Y'_a + V_{bn}Y'_b + V_{cn}Y'_c}{Y'_a + Y'_b + Y'_c}$$

where $Y'_x = \frac{1}{Z'_x}$.

3. Zero Neutral Current:

Without a neutral path, $I_{Nn} = 0$ strictly holds, which means:

$$I_{aA} + I_{bB} + I_{cC} = 0$$

EXAM PITFALL & CAUTION:

Even though $I_{Nn} = 0$ (no physical wire), V_{Nn} is NOT zero! The neutral point floats and shifts severely, subjecting some phases to dangerous overvoltages.

CH06-07 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 4 (12 HP Y-Motor)

3-Phase Motor Input Power, Efficiency & Line Current

PROBLEM / QUESTION:

A 3ϕ, 12-hp induction motor is connected in Y to a balanced $380\ \text{V}, 50\ \text{Hz}$ supply. The motor has efficiency $\eta = 90\%$ and $PF = 0.85$ lagging. Calculate: (1) Input electrical power P_{in}, (2) Line current I_L, (3) Total complex power S, (4) Equivalent impedance per phase Z_Y.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{out}} = 12 \times 746\ \text{W} = 8,952\ \text{W}$$

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta}$$

$$I_L = \frac{P_{\text{in}}}{\sqrt{3} V_L \cdot PF}$$

$$S = \frac{P_{\text{in}}}{PF} \angle \arccos(PF)$$

STEP-BY-STEP MODEL SOLUTION:

1. Convert output power to Watts:

$$P_{\text{out}} = 12 \times 746 = 8,952\ \text{W}$$

2. Input Electrical Real Power:

$$P_{\text{in}} = \frac{8,952}{0.90} = 9,946.67\ \text{W} \approx 9.95\ \text{kW}$$

3. Line Current (I_L):

$$I_L = \frac{9,946.67}{\sqrt{3} \times 380 \times 0.85} = \frac{9,946.67}{559.45} = 17.78\ \text{A}$$

4. Complex Power (S):

$$\theta = \arccos(0.85) = 31.79^\circ$$

$$|S| = \frac{9.946}{0.85} = 11.70\ \text{kVA}$$

$$S = 11.70 \angle 31.79^\circ\ \text{kVA} = 9.95\ \text{kW} + j6.16\ \text{kVAR}$$

5. Per-Phase Impedance (Z_Y):

$$V_p = \frac{380}{\sqrt{3}} = 219.39\ \text{V}$$

$$Z_Y = \frac{V_p}{I_L} = \frac{219.39}{17.78} \angle 31.79^\circ = 12.34 \angle 31.79^\circ\ \Omega = 10.49 + j6.50\ \Omega$$

($R = 10.49\ \Omega$, $L = \frac{6.50}{2\pi \times 50} = 20.7\ \text{mH}$)

EXAM PITFALL & CAUTION:

Remember: $1\ \text{hp} = 746\ \text{W}$. Horsepower is ALWAYS mechanical output power at the shaft (P_{out}), never electrical input power (P_{in})!

CH06-08 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 5 (10 HP Δ-Motor)

Delta-Connected 3-Phase Motor Analysis

PROBLEM / QUESTION:

A 3ϕ, 10-hp induction motor is connected in Δ to a $380\ \text{V}, 50\ \text{Hz}$ supply. $\eta = 85\%, PF = 0.80$ lagging. Find: (1) Input power P_{in}, (2) Line current I_L, (3) Phase winding current I_Δ, (4) Phase impedance Z_Δ.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{in}} = \frac{10 \times 746}{0.85}$$
$$I_L = \frac{P_{\text{in}}}{\sqrt{3} V_L PF}$$
$$I_\Delta = \frac{I_L}{\sqrt{3}}$$
$$Z_\Delta = \frac{V_L}{I_\Delta}$$

STEP-BY-STEP MODEL SOLUTION:

1. Input Power:

$$P_{\text{in}} = \frac{7,460\ \text{W}}{0.85} = 8,776.47\ \text{W} = 8.78\ \text{kW}$$

2. Line Current:

$$I_L = \frac{8,776.47}{\sqrt{3} \times 380 \times 0.80} = 16.67\ \text{A}$$

3. Delta Phase Current:
In Δ, line current divides: $I_\Delta = \frac{I_L}{\sqrt{3}} = \frac{16.67}{1.732} = 9.62\ \text{A}$

4. Impedance per phase (Z_Δ):
In Δ, $V_p = V_L = 380\ \text{V}$.

$$|Z_\Delta| = \frac{V_p}{I_\Delta} = \frac{380}{9.62} = 39.50\ \Omega$$

$$\theta = \arccos(0.8) = 36.87^\circ$$

$$Z_\Delta = 39.50 \angle 36.87^\circ\ \Omega = 31.60 + j23.70\ \Omega$$

EXAM PITFALL & CAUTION:

In a Delta load, $V_p = V_L = 380\ \text{V}$, but $I_p = I_L/\sqrt{3}$. Do not divide voltage by $\sqrt{3}$ for a Delta load!

Two-Wattmeter Method for 3-Phase Power Measurement

📌 **PROBLEM / QUESTION:**

In the two-wattmeter method measuring a balanced 3-phase load, state the equations for total real power (P_{total}), total reactive power (Q_{total}), and the load power factor angle (θ) in terms of wattmeter readings W_1 and W_2.

📐 **KEY FORMULAS & DEFINITIONS:**

$$P_{\text{total}} = W_1 + W_2$$
$$Q_{\text{total}} = \sqrt{3}(W_1 - W_2)$$
$$\tan\theta = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Total Active Power is simply the algebraic sum:

$$P = W_1 + W_2$$

2. Total Reactive Power is proportional to the difference:

$$Q = \sqrt{3}(W_1 - W_2)$$

3. Phase angle of the load:

$$\theta = \arctan\left(\sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}\right)$$

$$PF = \cos\theta$$

Special Cases:
- If $PF = 1.0$: $W_1 = W_2$
- If $PF = 0.5$: One wattmeter reads zero ($W_2 = 0$)
- If $PF < 0.5$: One wattmeter reads NEGATIVE!

⚠️ **EXAM PITFALL & CAUTION:**

When $PF < 0.5$, one wattmeter reads negative! On an analog meter, reverse the current coil connections to read the value, then treat that reading as a negative number in $W_1 + W_2$.

Power Factor Correction with Delta Capacitors

PROBLEM / QUESTION:

A balanced 3ϕ inductive load of $30\ \text{kW}$ operates at $380\ \text{V}, 50\ \text{Hz}$ with $PF_1 = 0.65$ lagging. What total reactive power Q_C must a capacitor bank supply to improve the power factor to $PF_2 = 0.95$ lagging, and what is the capacitance per phase if connected in Delta (C_Δ)?

KEY FORMULAS & DEFINITIONS:

$$Q_C = P(\tan\theta_1 - \tan\theta_2)$$

$$C_\Delta = \frac{Q_{C,\text{phase}}}{\omega V_L^2} = \frac{Q_C/3}{2\pi f V_L^2}$$

STEP-BY-STEP MODEL SOLUTION:

1. Find power factor angles:

- $\theta_1 = \arccos(0.65) = 49.46^\circ \Rightarrow \tan\theta_1 = 1.169$
- $\theta_2 = \arccos(0.95) = 18.19^\circ \Rightarrow \tan\theta_2 = 0.329$

2. Required 3-phase Reactive Power from Capacitors:

$$Q_C = 30\ \text{kW} \times (1.169 - 0.329) = 30 \times 0.840 = 25.20\ \text{kVAR}$$

3. Delta connection per-phase capacitance:

Each branch sees full line voltage $V_L = 380\ \text{V}$ and supplies $Q_{C1} = \frac{25{,}200}{3} = 8{,}400\ \text{VAR}$:

$$C_\Delta = \frac{8{,}400}{2\pi(50)(380)^2} = \frac{8{,}400}{314.16 \times 144{,}400} = 1.85 \times 10^{-4}\ \text{F} = 185\ \mu\text{F}$$

EXAM PITFALL & CAUTION:

Capacitors connected in Delta require 3 times LESS capacitance than Wye-connected capacitors ($C_\Delta = C_Y/3$) for the same reactive power, saving massive physical space and cost!

PART 2

Chapter 7: Electric Machinery & Transformers

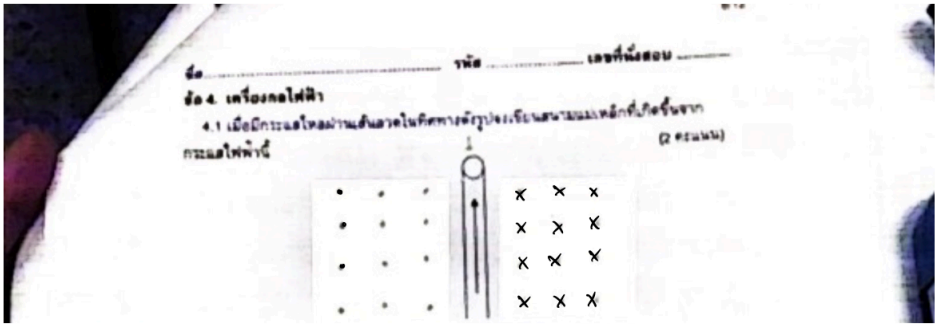
Right-Hand Rule, DC Shunt/Series Motors & Generators, Induction Motors, Slip, Power Transformers & Tests (27 Flashcards)

CH07-01 | Chapter 7 | Level 1 (Core Concept) | Final Exam Problem 4.1 (Page 28 - 2 Points)

Right-Hand Grip Rule for Magnetic Field around a Conductor

PROBLEM / QUESTION:

Current flows upward along a straight vertical wire as shown below. Using the right-hand grip rule, sketch/state the direction of the magnetic field ($\vec{B}$) on both sides of the wire.



รูปประกอบ: Exam Problem 4.1: Right-Hand Grip Rule

KEY FORMULAS & DEFINITIONS:

Thumb = Current (I) Fingers curl = Magnetic field ($\vec{B}$)

STEP-BY-STEP MODEL SOLUTION:

1. Align the right thumb with current flow pointing UPWARD.
2. Curled fingers point:
- Out of the page on the LEFT side (represented by dots •)
- Into the page on the RIGHT side (represented by crosses ×)

EXAM PITFALL & CAUTION:

Always use the RIGHT hand! Left-hand grip reverses all field directions.

CH07-02 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.2 (Page 28 - 3 Points)

DC Motor Magnetic Force on Wire (Lorentz Force Rule)

📌 **PROBLEM / QUESTION:**

In the DC motor setup shown below, the upper permanent magnet pole is North (N) and the lower is South (S). Current is applied to the loop. State: (1) Current direction, (2) Direction of magnetic field $\vec{B}$, (3) Direction of resulting force $\vec{F}$ on each coil side.



รูปประกอบ: Exam Problem 4.2: DC Motor Magnetic Field & Force (Lorentz)

📐 **KEY FORMULAS & DEFINITIONS:**

$$\vec{F} = I(\vec{L} \times \vec{B})$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Magnetic field $\vec{B}$ points from North to South (DOWNWARD).
2. Current flows from (+) terminal through the coil to (-) terminal.
3. By Fleming's Left-Hand Rule (or $\vec{L} \times \vec{B}$):

- The two opposite coil sides experience opposing forces forming a clockwise or counterclockwise torque that rotates the rotor shaft.

⚠️ **EXAM PITFALL & CAUTION:**

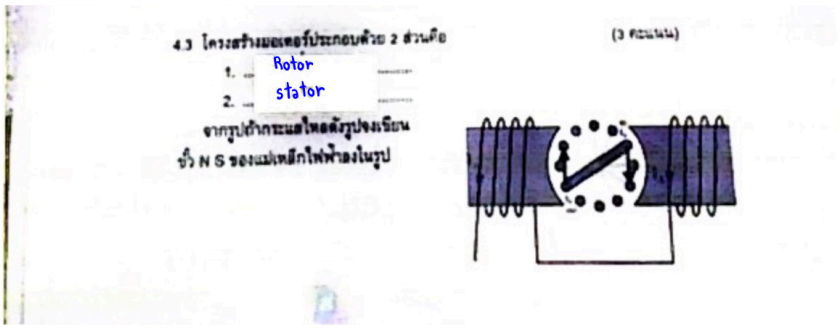
For motors, use Fleming's LEFT-hand rule. For generators, use Fleming's RIGHT-hand rule.

CH07-03 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.3 (Page 28 - 3 Points)

DC Machine Stator vs Rotor & Electromagnet Pole Polarities

📌 **PROBLEM / QUESTION:**

What are the two primary structural parts of an electric motor? For the electromagnet core shown below, identify the magnetic poles (N and S) established by the coil current.



รูปประกอบ: Exam Problem 4.3: Motor Magnetic Poles (N & S)

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Right-Hand Solenoid Rule: Fingers} \rightarrow \vec{I}_{\text{curl}}, \quad \text{Thumb} \rightarrow \text{North Pole (N)}$$

$$\vec{F} = I(\vec{L} \times \vec{B})$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. The two essential parts are:
- Stator (ส่วนที่อยู่กับที่): Stationary field frame providing magnetic flux $\vec{B}$.
- Rotor / Armature (ส่วนที่หมุนเคลื่อนที่): Rotating shaft carrying current coils that generate torque $\vec{\tau}$.

2. Solenoid Pole Identification:
Using the right-hand coil rule, curl four fingers in the direction current wraps around the iron pole pieces. The thumb points toward the North pole (N) where magnetic flux emerges:

$$\text{Current direction: } \vec{I} \implies \text{Magnetic dipole: } \vec{m} \parallel \text{Thumb (N)}$$

⚠️ **EXAM PITFALL & CAUTION:**

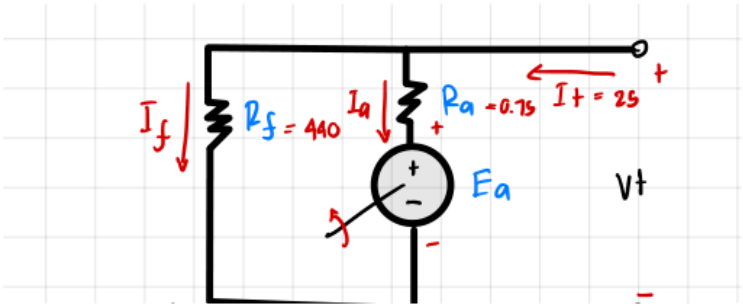
Do not confuse the armature core with the stator frame. Stator provides the stationary B-field; rotor carries the torque-producing coils.

CH07-04 | Chapter 7 | Level 1 (Core Concept) | Final Exam Problem 4.4 (Page 27 - 8 Points)

3-Phase Induction Motor & Squirrel-Cage Rotor Operation

PROBLEM / QUESTION:

Referring to the 3-phase induction motor diagram below: (1) What is a Squirrel-Cage Rotor and how does it work? (2) Explain the operating principle of a 3-phase induction motor. (3) How do you reverse its direction of rotation?



รูปประกอบ: Homework 7 Problem 4: DC Shunt Motor Equivalent Circuit

KEY FORMULAS & DEFINITIONS:

$$n_s = \frac{120f}{P}$$

$$s = \frac{n_s - n_r}{n_s}$$

STEP-BY-STEP MODEL SOLUTION:

1. Squirrel-Cage Rotor (โรเตอร์แบบกรงกระรอก):
Consists of conducting aluminum/copper bars embedded in laminated iron core, short-circuited at both ends by end-rings. When the rotating magnetic field sweeps across the bars, it induces currents producing torque.

2. Operating Principle:
3-phase stator currents produce a Constant-Magnitude Rotating Magnetic Field (RMF) spinning at synchronous speed $n_s = 120f/P$. By Faraday-Lenz law, this field cuts rotor bars, inducing voltage and rotor currents. The interaction generates Lorentz torque, pulling the rotor in the direction of the rotating field.

3. Direction Reversal:
Swap ANY TWO supply phase lines (e.g., swap A and B, keeping C fixed). This reverses the phase sequence and rotates the RMF in the opposite direction.

EXAM PITFALL & CAUTION:

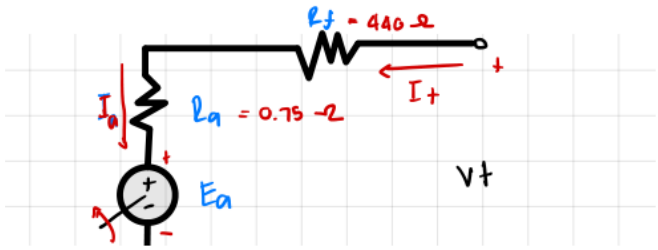
The rotor can NEVER reach synchronous speed ($n_r < n_s$). If $n_r = n_s$, relative velocity is zero, no voltage is induced, and torque drops to zero!

CH07-05 Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.5 (Page 26 - 4 Points)

DC Generator Principle & Commutator Rectification

📌 **PROBLEM / QUESTION:**

What machine is shown in the diagram below and how does it work? Explain how the split-ring commutator converts induced AC into pulsating DC at the output terminals.



รูปประกอบ: Homework 7 Problem 5: DC Series Motor Equivalent Circuit

📐 **KEY FORMULAS & DEFINITIONS:**

$$e = B\ell v \sin\theta$$

$$E_a = k_a \phi \omega_m$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Identification:
Single-phase DC Generator with split-ring commutator.

2. Principle of Operation:
Mechanical power drives the coil rotating in the magnetic field between permanent magnet poles. By Faraday's Law, an alternating electromotive force ($e = B\ell v$) is induced inside the rotating coil.

3. Commutator Action:
The split-ring commutator mechanically reverses coil connections to the external stationary carbon brushes every half revolution, rectifying the internal alternating AC into unidirectional pulsating DC.

⚠️ **EXAM PITFALL & CAUTION:**

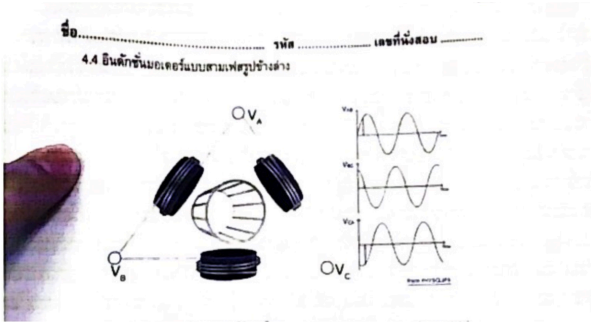
Inside ANY rotating DC armature, the induced voltage is ALWAYS AC! The commutator is what converts internal AC into external DC.

CH07-06 | Chapter 7 | Level 2 (Exam / Calc) | HW Ch.7 Problem 4 & Talui Homework Page 1

DC Shunt Motor Equivalent Circuit, Speed & Torque

PROBLEM / QUESTION:

A 2-pole, $2.5\text{-hp}, 100\ \text{VDC}$ shunt motor has design constant $k_a = 153$. At full load, $I_t = 25.0\ \text{A}, R_a = 0.75\ \Omega, \phi = 10.0\ \text{mWb}, R_f = 440\ \Omega$. Draw the equivalent circuit, and calculate: (1) Back EMF E_a, (2) Rotational speed n [rpm], (3) Developed torque τ_d [N·m].



รูปประกอบ: Exam Problem 4.4: 3-Phase Induction Motor (Squirrel Cage)

KEY FORMULAS & DEFINITIONS:

$$I_f = \frac{V_t}{R_f}, \quad I_a = I_t - I_f$$
$$E_a = V_t - I_a R_a = k_a \phi \omega_m$$
$$\omega_m = \frac{2\pi n}{60}, \quad \tau_d = k_a \phi I_a$$

STEP-BY-STEP MODEL SOLUTION:

1. Field current & Armature current:

$$I_f = \frac{V_t}{R_f} = \frac{100}{440} = 0.23\ \text{A}$$

$$I_a = I_t - I_f = 25.0 - 0.23 = 24.77\ \text{A}$$

2. Back EMF (E_a):

$$E_a = V_t - I_a R_a = 100 - (24.77 \times 0.75) = 100 - 18.58 = 81.42\ \text{V}$$

3. Speed (n):

$$E_a = k_a \phi \omega_m \Rightarrow 81.42 = 153 \times (10 \times 10^{-3}) \times \omega_m$$

$$\omega_m = \frac{81.42}{1.53} = 53.22\ \text{rad/s}$$

$$n = \frac{60 \times \omega_m}{2\pi} = \frac{60 \times 53.22}{2\pi} = 508.17\ \text{rpm}$$

4. Developed Torque (τ_d):

$$\tau_d = k_a \phi I_a = 153 \times (10 \times 10^{-3}) \times 24.77 = 37.90\ \text{N} \cdot \text{m}$$

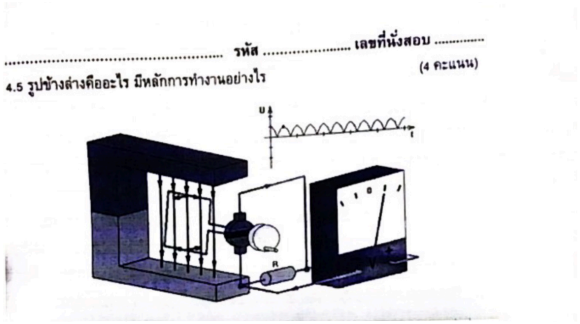
EXAM PITFALL & CAUTION:

In a Shunt motor, $I_t = I_a + I_f$. Do NOT use total line current I_t in $V_t - IR_a$; only I_a passes through the armature resistance!

DC Series Motor Circuit & Terminal Voltage Calculation

PROBLEM / QUESTION:

A 2-pole, 2.5-hp DC Series motor has $k_a = 153, R_a = 0.75\ \Omega, R_f = 440\ \Omega, \phi = 10.0\ \text{mWb}$. Find: (a) Terminal voltage V_{t1} giving the SAME SPEED ($n = 508.17$ rpm) as Problem 4, (b) Terminal voltage V_{t2} giving the SAME TORQUE ($\tau_d = 37.9\ \text{N} \cdot \text{m}$) as Problem 4.



รูปประกอบ: Exam Problem 4.5: DC Generator Principle & Split-Ring Commutator

KEY FORMULAS & DEFINITIONS:

Series connection: $I_t = I_a = I_f$

$$V_t = E_a + I_a(R_a + R_f)$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Same Speed ($n = 508.17$ rpm):

$E_a = k_a \phi \omega_m = 81.42\ \text{V}$.

From power $P_e = E_a I_a = 1,865\ \text{W} \Rightarrow I_a = \frac{1,865}{81.42} = 22.91\ \text{A}$.

$$V_{t1} = E_a + I_a(R_a + R_f) = 81.42 + 22.91(0.75 + 440) = 81.42 + 10,097.6 = 10.18\ \text{kV}$$

2. Part (b) Same Torque ($\tau_d = 37.9\ \text{N} \cdot \text{m}$):

From $\tau_d = k_a \phi I_a \Rightarrow I_a = 24.77\ \text{A}$.

$P_e = E_a I_a = 1,865\ \text{W} \Rightarrow E_a = \frac{1,865}{24.77} = 75.30\ \text{V}$.

$$V_{t2} = E_a + I_a(R_a + R_f) = 75.30 + 24.77(440.75) = 11.00\ \text{kV}$$

EXAM PITFALL & CAUTION:

In a Series motor, the field winding is in series with the armature ($I_a = I_f = I_t$). The total circuit resistance is $R_a + R_f$.

CH07-08 | Chapter 7 | Level 2 (Exam / Calc) | FunE-final.pdf Page 5 (Example 3)

DC Shunt Motor Speed Control via Armature Resistance

PROBLEM / QUESTION:

A $220\ \text{V}$ DC shunt motor has $R_a = 0.5\ \Omega, R_f = 110\ \Omega$. At full load, $I_t = 42\ \text{A}$ and speed $n_1 = 1,200\ \text{rpm}$. If an external resistor $R_{\text{ext}} = 1.0\ \Omega$ is inserted in series with the armature while torque remains constant, calculate the new motor speed n_2.

KEY FORMULAS & DEFINITIONS:

$$E_{a1} = V_t - I_{a1}R_a$$
$$E_{a2} = V_t - I_{a2}(R_a + R_{\text{ext}})$$
$$\frac{n_2}{n_1} = \frac{E_{a2}}{E_{a1}} \quad (\text{when } \phi \text{ is constant})$$

STEP-BY-STEP MODEL SOLUTION:

1. Initial condition:

$$I_f = \frac{220}{110} = 2.0\ \text{A}$$

$$I_{a1} = 42 - 2 = 40.0\ \text{A}$$

$$E_{a1} = 220 - (40 \times 0.5) = 220 - 20 = 200\ \text{V}$$

2. New condition with $R_{\text{ext}} = 1.0\ \Omega$:
Constant torque and constant flux $\Rightarrow I_{a2} = I_{a1} = 40.0\ \text{A}$.

$$E_{a2} = 220 - 40(0.5 + 1.0) = 220 - 60 = 160\ \text{V}$$

3. Calculate new speed:

$$\frac{n_2}{1200} = \frac{160}{200} = 0.80 \Rightarrow n_2 = 1,200 \times 0.80 = 960\ \text{rpm}$$

EXAM PITFALL & CAUTION:

Adding armature resistance can ONLY reduce speed below base speed ($n < n_{\text{base}}$). To run above base speed, field weakening (reducing I_f) is required.

CH07-09 | Chapter 7 | Level 2 (Exam / Calc) | FunE-final.pdf Page 4 (Example 2)

DC Shunt Generator Voltage Regulation & KVL

PROBLEM / QUESTION:

A DC shunt generator driven at $1,800$ rpm produces no-load terminal voltage $V_t = 115$ V. When connected to a load, $I_a = 40$ A and terminal voltage drops to $V_t = 100$ V. Find: (1) Armature resistance R_a, (2) Voltage regulation ($VR\%$), (3) Counter-torque τ_d at full load if $\phi = 10$ mWb.

KEY FORMULAS & DEFINITIONS:

Generator KVL:

$$E_a = V_t + I_a R_a$$

$$VR\% = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

STEP-BY-STEP MODEL SOLUTION:

1. Armature Resistance (R_a):
At no load ($I_a = 0$), $E_a = V_{NL} = 115$ V.
At full load: $115 = 100 + (40 \times R_a) \Rightarrow R_a = \frac{15}{40} = 0.375\ \Omega$.

2. Voltage Regulation:

$$VR\% = \frac{115 - 100}{100} \times 100\% = 15.0\%$$

3. Counter Torque (τ_d):

$$\omega_m = \frac{2\pi(1,800)}{60} = 60\pi = 188.5\ \text{rad/s}$$

$$P_e = E_a I_a = 115 \times 40 = 4,600\ \text{W}$$

$$\tau_d = \frac{P_e}{\omega_m} = \frac{4,600}{188.5} = 24.40\ \text{N} \cdot \text{m}$$

EXAM PITFALL & CAUTION:

Generator KVL has a PLUS sign ($E_a = V_t + I_a R_a$), while Motor KVL has a MINUS sign ($E_a = V_t - I_a R_a$)! In a generator, internal generated EMF is higher than terminal voltage.

CH07-10 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Final Exam

Induction Motor Synchronous Speed, Rotor Speed & Slip

PROBLEM / QUESTION:

A 4-pole, $50\ \text{Hz}$ three-phase induction motor runs at $1,440\ \text{rpm}$ at full load. Calculate: (1) Synchronous speed n_s, (2) Fractional slip s and percent slip $s\%$, (3) Rotor current frequency f_r, (4) Rotor speed at locked rotor ($s = 1$).

KEY FORMULAS & DEFINITIONS:

$$n_s = \frac{120f}{P}$$
$$s = \frac{n_s - n_r}{n_s}$$
$$f_r = sf$$

STEP-BY-STEP MODEL SOLUTION:

1. Synchronous Speed:

$$n_s = \frac{120 \times 50}{4} = 1,500\ \text{rpm}$$

2. Slip (s):

$$s = \frac{1,500 - 1,440}{1,500} = \frac{60}{1,500} = 0.04 = 4.0\%$$

3. Rotor Current Frequency (f_r):

$$f_r = s \times f = 0.04 \times 50\ \text{Hz} = 2.0\ \text{Hz}$$

4. At Locked-Rotor:
$n_r = 0$, slip $s = 1.0$, and rotor frequency equals line frequency ($f_r = 50\ \text{Hz}$).

EXAM PITFALL & CAUTION:

Rotor electrical frequency is sf, NOT f! At normal running speed, slip is small ($2 - 5\%$), so rotor frequency is tiny ($1 - 2.5\ \text{Hz}$).

CH07-11 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Final Exam

Induction Motor Power Flow: Air-Gap, Copper Loss & Mech Power

PROBLEM / QUESTION:

State the universal proportional power relationship between Air-gap Power (P_{ag}), Rotor Copper Loss (P_{rcu}), and Converted Mechanical Power (P_{conv}) in an induction motor.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{ag}} : P_{\text{rcu}} : P_{\text{conv}} = 1 : s : (1-s)$$

$$P_{\text{rcu}} = sP_{\text{ag}}$$

$$P_{\text{conv}} = (1-s)P_{\text{ag}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Total air-gap power crosses the stator-rotor gap: P_{ag}.
2. Fraction s is dissipated as rotor I^2R heat: $P_{\text{rcu}} = sP_{\text{ag}}$.
3. Fraction $(1-s)$ is converted into mechanical power: $P_{\text{conv}} = (1-s)P_{\text{ag}}$.
4. Internal developed torque is: $\tau_d = \frac{P_{\text{ag}}}{\omega_s} = \frac{P_{\text{conv}}}{\omega_m}$.

EXAM PITFALL & CAUTION:

Rotor copper loss is DIRECTLY proportional to slip! High slip means high rotor heating and very poor electrical efficiency.

CH07-12 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Final Exam

Synchronous Motor Operation, V-Curve & Power Factor Correction

PROBLEM / QUESTION:

Why does a Synchronous Motor rotate at exactly synchronous speed with zero slip ($s = 0$)? How does varying its DC field excitation current (I_f) allow it to act as a Synchronous Condenser to correct facility power factor?

KEY FORMULAS & DEFINITIONS:

$$n_r = n_s = \frac{120f}{P}$$

Normal excitation: $PF = 1.0$ Over-excitation: Leading PF (supplies Q) Under-excitation: Lagging PF (draws Q)

STEP-BY-STEP MODEL SOLUTION:

1. Constant Synchronous Speed:
The rotor's DC-excited poles lock magnetically into synchronism with the stator's rotating magnetic field. As long as load does not exceed pull-out torque, it runs at precisely n_s.

2. V-Curve & Power Factor Control:
- Under-excited (I_f small): Motor draws lagging reactive power ($Q > 0$).
- Normal excitation: Motor operates at unity power factor ($PF = 1.0$), minimum stator current.
- Over-excited (I_f large): Motor draws leading reactive power (acts as a capacitor, supplying VARs to the grid to correct factory power factor).

EXAM PITFALL & CAUTION:

Synchronous motors are NOT inherently self-starting from a dead stop because the stator RMF moves past stationary rotor poles too fast for inertia to catch up. They require damper (amortisseur) windings to start as an induction motor first.

CH07-13 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Review

DC Motor Starting Current & Starting Resistor Design

📌 **PROBLEM / QUESTION:**

Why is the starting current of a DC motor dangerously high (10 to 20 times rated current) if started directly on full voltage? How does an external starting resistor protect the motor?

📐 **KEY FORMULAS & DEFINITIONS:**

At start ($n = 0$): $E_a = 0$

$$I_{\text{start}} = \frac{V_t}{R_a}$$

With starter:

$$I_{\text{start}} = \frac{V_t}{R_a + R_{\text{start}}}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Why Starting Current is Enormous:

At standstill ($n = 0$), back EMF $E_a = k_a \phi \omega_m = 0$ V. Because armature resistance R_a is designed to be tiny ($< 0.5\ \Omega$) for high efficiency:

$$I_{\text{start}} = \frac{V_t - 0}{R_a} = \frac{220\ \text{V}}{0.5\ \Omega} = 440\ \text{A!}$$

This burns the armature coils and melts commutator brushes.

2. Protection via Starting Resistor:

A multi-step starting resistance R_{start} is placed in series with the armature to limit starting current to $1.5 - 2.0 \times I_{\text{rated}}$. As the motor accelerates and builds up E_a, resistance steps are cut out.

⚠️ **EXAM PITFALL & CAUTION:**

Never start a DC motor under full voltage without starting resistance (except tiny sub-fractional HP motors).

CH07-14 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Exam Pitfalls

Danger of Open Field Circuit in DC Shunt Motor

📌 **PROBLEM / QUESTION:**

What happens if the shunt field circuit of a running DC shunt motor accidentally breaks or becomes open-circuited ($I_f = 0$)? Explain the physical mechanism.

📐 **KEY FORMULAS & DEFINITIONS:**

$$n = \frac{V_t - I_a R_a}{k_a \phi}$$

As $\phi \to \phi_{\text{residual}} \approx 0 \Rightarrow n \to \infty$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Field current drops to zero ($I_f = 0$), so magnetic flux drops to residual magnetism ($\phi \to \phi_{\text{residual}} \approx 0$).
2. Back EMF collapses: $E_a \approx 0$, causing armature current $I_a = \frac{V_t - E_a}{R_a}$ to soar to destructive levels.
3. Because $n \propto \frac{1}{\phi}$, the motor accelerates uncontrollably to catastrophic runaway speed ($n \to \infty$), resulting in mechanical disintegration (throwing armature windings and bearings exploding).

⚠️ **EXAM PITFALL & CAUTION:**

Never install a fuse or switch in the field winding of a DC shunt motor! An open field causes motor destruction.

CH07-15 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Safety Rules

DC Series Motor Prohibition Against No-Load Operation

📌 **PROBLEM / QUESTION:**

Why is it strictly forbidden to run a DC Series Motor without a mechanical load (or with a belt drive that could snap)?

📐 **KEY FORMULAS & DEFINITIONS:**

Series motor: $\phi \propto I_a$

$$n \propto \frac{V_t - I_a(R_a + R_f)}{k_a \phi} \approx \frac{V_t}{k' I_a}$$

As $I_a \to 0 \Rightarrow n \to \infty$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. In a series motor, field current equals armature current ($I_f = I_a$).
2. At no load, torque required is near zero, so armature current drops close to zero ($I_a \to 0$).
3. This causes magnetic flux to drop to residual levels ($\phi \to 0$).
4. Speed $n \propto \frac{1}{\phi} \propto \frac{1}{I_a}$ races towards dangerous infinity ($n \to \infty$), destroying the machine by centrifugal force.

⚠️ **EXAM PITFALL & CAUTION:**

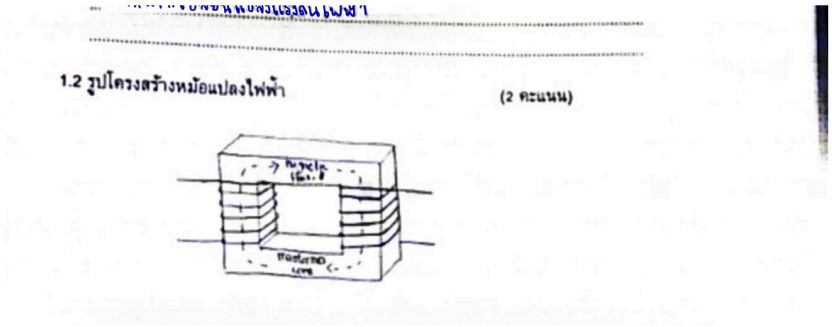
Series motors MUST ALWAYS be solidly coupled (gears/direct shaft) to their load, NEVER belt-driven!

CH07-16 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 1.1 & 1.2 (Page 32 - 4 Points)

Transformer Definition, Purpose & Structure

📌 **PROBLEM / QUESTION:**

What is an electrical transformer? What is its primary function and structure? Draw/describe its essential construction.



รูปประกอบ: Exam Problem 1.2: Transformer Core & Dual Windings

📐 **KEY FORMULAS & DEFINITIONS:**

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = a, \quad \frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{1}{a}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Definition & Function (ข้อ 1.1):
- A static electrical machine with NO moving parts that transfers electrical energy between circuits through electromagnetic induction.
- Purpose: Step-up (เพิ่มแรงดัน) or Step-down (ลดแรงดัน) AC voltage levels while keeping frequency constant.

2. Structure (ข้อ 1.2):
- Laminated ferromagnetic core (แกนเหล็กแผ่นซิลิคอนบางเรียงซ้อนกันเพื่อลดกระแสไหลวน Eddy Current)
- Primary winding (ขดลวดปฐมภูมิ N_1 ต่อเข้าแหล่งจ่าย)
- Secondary winding (ขดลวดทุติยภูมิ N_2 ต่อเข้าโหลด)

⚠️ **EXAM PITFALL & CAUTION:**

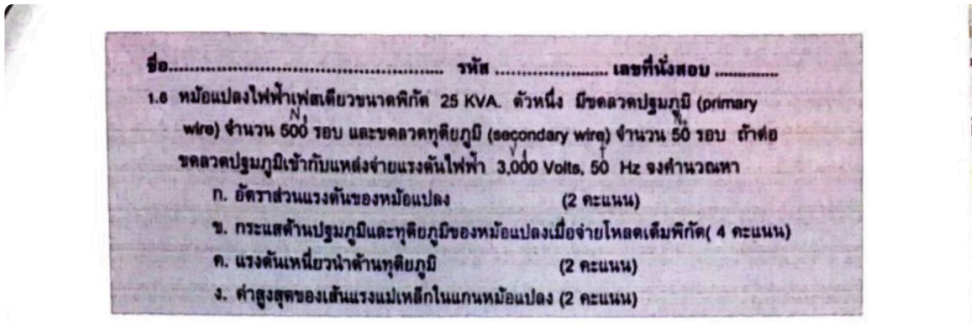
Transformers operate ONLY on Alternating Current (AC)! Never connect a transformer to a DC supply, as absence of changing flux ($d\phi/dt = 0$) induces zero back-EMF, causing excessive current that burns the primary winding.

CH07-17 | Chapter 7 | Level 2 (Exam / Calc) | Final Exam Problem 1.6 (Page 31 - 10 Points)

Single Phase 25 kVA Transformer Full Exam Problem

PROBLEM / QUESTION:

A single-phase transformer rated $25\ \text{kVA}$ has $N_1 = 500$ turns and $N_2 = 50$ turns. Primary winding is connected to $V_1 = 3,000\ \text{V}, 50\ \text{Hz}$. Calculate: (a) Turns ratio a, (b) Rated primary & secondary currents (I_1, I_2), (c) Induced secondary voltage V_2, (d) Maximum core flux Φ_m.


ชื่อ..... รหัส ..... เลขที่นั่งสอบ .....

1.6 หม้อแปลงไฟฟ้าเฟสเดียวขนาดพิกัด 25 KVA. ตัวหนึ่ง มีขดลวดปฐมภูมิ (primary wire) จำนวน 500 รอบ และขดลวดทุติยภูมิ (secondary wire) จำนวน 50 รอบ ถ้าต่อขดลวดปฐมภูมิเข้ากับแหล่งจ่ายแรงดันไฟฟ้า 3,000 Volts, 50 Hz จงคำนวณหา

ก. อัตราส่วนแรงดันของหม้อแปลง (2 คะแนน)

ข. กระแสด้านปฐมภูมิและทุติยภูมิของหม้อแปลงเมื่อจ่ายโหลดเต็มพิกัด(4 คะแนน)

ค. แรงดันเหนี่ยวนำด้านทุติยภูมิ (2 คะแนน)

ง. ค่าสูงสุดของเส้นแรงแม่เหล็กในแกนหม้อแปลง (2 คะแนน)

รูปประกอบ: Exam Problem 1.6: 25 kVA Transformer Full Calculation

KEY FORMULAS & DEFINITIONS:

$$a = \frac{N_1}{N_2}, \quad I_1 = \frac{S}{V_1}, \quad I_2 = \frac{S}{V_2}$$
$$V_2 = \frac{V_1}{a}, \quad V_1 = 4.44 f N_1 \Phi_m$$

STEP-BY-STEP MODEL SOLUTION:

1. (ก) Turns ratio (a):

$$a = \frac{N_1}{N_2} = \frac{500}{50} = 10$$

2. (ข) Rated currents:

$$I_1 = \frac{S}{V_1} = \frac{25,000}{3,000} = 8.33\ \text{A} = \frac{25}{3}\ \text{A}$$

$$I_2 = aI_1 = 10 \times 8.33 = 83.33\ \text{A} = \frac{250}{3}\ \text{A}$$

3. (ค) Secondary induced voltage (V_2):

$$V_2 = \frac{V_1}{a} = \frac{3,000}{10} = 300\ \text{V}$$

4. (ง) Maximum core flux (Φ_m):

$$V_1 = 4.44 f N_1 \Phi_m \Rightarrow 3,000 = 4.44(50)(500)\Phi_m$$

$$\Phi_m = \frac{3,000}{111,000} = 0.027\ \text{Wb} = 27.0\ \text{mWb}$$

EXAM PITFALL & CAUTION:

The constant 4.44 comes from $2\pi/\sqrt{2} \approx 4.44288$. Flux Φ_m is in Webers (Wb). Remember: V_1 uses N_1; V_2 uses N_2.

CH07-18 | Chapter 7 | Level 2 (Exam / Calc) | HW Ch.8 Problem 2 (Page 1)

Transformer: Turns Ratio & Winding Calculations

PROBLEM / QUESTION:

The primary and secondary voltages of a single-phase transformer are $V_1 = 140\ \text{V}$ and $V_2 = 80\ \text{V}$. Primary current is $I_1 = 8\ \text{A}$. Secondary winding has $N_2 = 800$ turns. Find: (1) Primary winding turns N_1, (2) Secondary current I_2, (3) Turns ratio a.

KEY FORMULAS & DEFINITIONS:

$$a = \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Turns Ratio (a):

$$a = \frac{V_1}{V_2} = \frac{140}{80} = 1.75$$

2. Primary turns (N_1):

$$N_1 = aN_2 = 1.75 \times 800 = 1,400\ \text{turns}$$

3. Secondary current (I_2):

$$I_2 = aI_1 = 1.75 \times 8\ \text{A} = 14.0\ \text{A}$$

EXAM PITFALL & CAUTION:

Voltage and turns are directly proportional ($V_1/V_2 = N_1/N_2$), but current is INVERSELY proportional ($I_1/I_2 = N_2/N_1$).

CH07-19 | Chapter 7 | Level 2 (Exam / Calc) | HW Ch.8 Problem 3 (Page 1)

200 kVA Transformer Voltage, Current & Core Flux

PROBLEM / QUESTION:

A single-phase transformer rated $200\ \text{kVA}, 2200\ \text{V}/220\ \text{V}, 50\ \text{Hz}$ has primary winding $N_1 = 200$ turns. Find: (1) Rated primary and secondary currents (I_1, I_2), (2) Secondary turns N_2, (3) Maximum magnetic core flux Φ_m.

KEY FORMULAS & DEFINITIONS:

$$I_1 = \frac{S}{V_1}, \quad I_2 = \frac{S}{V_2}$$
$$N_2 = \frac{V_2}{V_1} N_1, \quad \Phi_m = \frac{V_1}{4.44 f N_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Rated Currents:

$$I_1 = \frac{200 \times 10^3}{2,200} = 90.91\ \text{A}$$

$$I_2 = \frac{200 \times 10^3}{220} = 909.09\ \text{A}$$

2. Secondary turns (N_2):

$$a = \frac{2,200}{220} = 10 \Rightarrow N_2 = \frac{N_1}{10} = \frac{200}{10} = 20\ \text{turns}$$

3. Maximum flux (Φ_m):

$$\Phi_m = \frac{2,200}{4.44 \times 50 \times 200} = \frac{2,200}{44,400} = 0.0495\ \text{Wb} = 49.5\ \text{mWb}$$

EXAM PITFALL & CAUTION:

Always double check: $V_2 = 4.44 f N_2 \Phi_m$ must yield the EXACT same flux value: $220/(4.44 \times 50 \times 20) = 49.5\ \text{mWb}$.

CH07-20 | Chapter 7 | Level 2 (Exam / Calc) | HW Ch.8 Problem 4.4 (Page 1)

Transformer DC Connection Danger (Why Transformers Cannot Run on DC)

PROBLEM / QUESTION:

A $25\ \text{kVA}, 500/40\ \text{turns}$ transformer is supplied at primary by $300\ \text{V}, 50\ \text{Hz}$. (1) If we change to connect a DC voltage of $200\ \text{V}$ to the primary winding, what will the secondary current I_2 be? (2) What happens physically to the transformer?

KEY FORMULAS & DEFINITIONS:

$$e_2 = -N_2 \frac{d\Phi}{dt} = 0 \quad (\text{for steady DC})$$

STEP-BY-STEP MODEL SOLUTION:

1. Secondary current (I_2):

$$I_2 = 0\ \text{A}$$

Because DC voltage produces constant, steady magnetic flux ($\frac{d\Phi}{dt} = 0$), no voltage is induced in the secondary winding ($e_2 = 0\ \text{V}$), so $I_2 = 0\ \text{A}$.

2. Physical Consequence to Transformer:
With no alternating flux, there is zero back-EMF ($e_1 = 0$). Primary current is limited only by tiny copper resistance R_1 ($< 0.1\ \Omega$):

$$I_{1,\text{DC}} = \frac{200\ \text{V}}{R_1} \gg 1,000\ \text{A}!$$

The primary winding will overheat instantly, burn its enamel insulation, and catch fire!

EXAM PITFALL & CAUTION:

Classic trick question! Never write $I_2 = 200/a$. For DC, induced secondary voltage and current are ZERO, while primary burns.

CH07-21 | Chapter 7 | Level 1 (Core Concept) | Final Exam Problem 1.4 (Page 32 - 2 Points)

Ideal Transformer 4 Fundamental Hypotheses

PROBLEM / QUESTION:

State the 4 fundamental properties/assumptions that define an Ideal Transformer (หม้อแปลงไฟฟ้าในอุดมคติ).

KEY FORMULAS & DEFINITIONS:

Ideal: $R_1 = R_2 = 0, \quad \mu_r = \infty, \quad P_{\text{loss}} = 0, \quad \Phi_{\text{leakage}} = 0$

STEP-BY-STEP MODEL SOLUTION:

1. Zero Winding Resistance: $R_1 = 0, R_2 = 0$ (No I^2R copper losses).
2. Infinite Core Permeability: $\mu_r \to \infty$, so zero reluctance and zero magnetizing current needed ($I_m = 0$).
3. Zero Core Losses: No hysteresis loss and no eddy current loss ($P_{\text{core}} = 0$).
4. Zero Leakage Flux: All magnetic flux is perfectly confined within the core and links both windings completely ($X_1 = X_2 = 0$).

EXAM PITFALL & CAUTION:

Real transformers have losses (efficiency $95 - 99\%$). An ideal transformer has 100% efficiency, so $S_{\text{in}} = S_{\text{out}}$ and $P_{\text{in}} = P_{\text{out}}$.

CH07-22 | Chapter 7 | Level 1 (Core Concept) | Final Exam Problem 1.5 (Page 32 - 2 Points)

Transformer Losses: Core Loss vs Copper Loss

PROBLEM / QUESTION:

What types of power losses occur in a real transformer? Contrast Core Loss (Iron Loss) with Copper Loss (Winding Loss) regarding their cause and load dependency.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{total loss}} = P_{\text{core}} + P_{\text{cu}}$$
$$P_{\text{core}} = P_h + P_e \quad (\text{constant } 24/7)$$
$$P_{\text{cu}} = I_1^2 R_1 + I_2^2 R_2 \propto I_L^2$$

STEP-BY-STEP MODEL SOLUTION:

1. Core Loss (Iron Loss / การสูญเสียในแกนเหล็ก):
- Hysteresis loss ($P_h \propto f B_{\text{max}}^{1.6}$) + Eddy current loss ($P_e \propto f^2 B_{\text{max}}^2 t^2$).
- Constant 24/7: Depends only on applied voltage and frequency; independent of load current!

2. Copper Loss (I^2R Loss / การสูญเสียในขดลวดทองแดง):
- Caused by Joule heating in primary and secondary winding resistances.
- Variable: Proportional to the square of load current ($P_{\text{cu}} \propto I_L^2$). Zero at no load!

EXAM PITFALL & CAUTION:

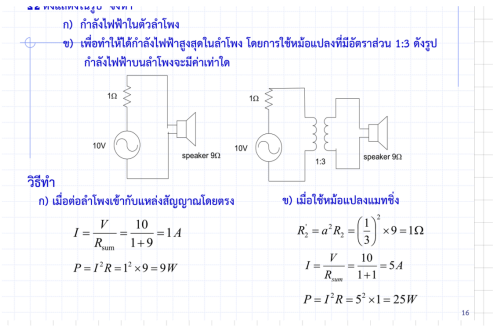
Maximum transformer efficiency occurs when Variable Loss equals Constant Loss: $P_{\text{cu}} = P_{\text{core}}$!

CH07-23 | Chapter 7 | Level 2 (Exam / Calc) | Ch07Transformer.pdf Example 1 (Page 18) & FunE P.7

Audio Transformer Impedance Matching for Maximum Power Transfer

PROBLEM / QUESTION:

A $10\ \text{V}$ audio source with internal source resistance $R_s = 1\ \Omega$ is connected to a speaker load $R_L = 9\ \Omega$. (a) Find power to the speaker without transformer. (b) If an ideal matching transformer with turns ratio $a = 1/3$ is inserted, find the new power transferred to the speaker.



รูปประกอบ: Audio Transformer Maximum Power Transfer Impedance Matching

KEY FORMULAS & DEFINITIONS:

$$R_L' = a^2 R_L = \left(\frac{N_1}{N_2}\right)^2 R_L$$

Maximum Power Transfer when $R_L' = R_s$

STEP-BY-STEP MODEL SOLUTION:

1. Without Transformer:

$$I = \frac{10}{1+9} = 1.0\ \text{A}$$

$$P_{\text{speaker}} = I^2 R_L = (1.0)^2 \times 9 = 9.0\ \text{W}$$

2. With Transformer ($a = 1/3$):

Reflected load resistance seen by source:

$$R_L' = a^2 R_L = \left(\frac{1}{3}\right)^2 \times 9 = \frac{1}{9} \times 9 = 1.0\ \Omega$$

Notice $R_L' = R_s = 1.0\ \Omega$ (perfect Maximum Power Transfer match!).

New primary current:

$$I_1 = \frac{10}{1+1} = 5.0\ \text{A}$$

Power delivered to reflected load (and thus speaker):

$$P_{\text{speaker}} = I_1^2 R_L' = (5.0)^2 \times 1.0 = 25.0\ \text{W}$$

Power increases dramatically from $9\ \text{W}$ to $25\ \text{W}$!

EXAM PITFALL & CAUTION:

Reflected impedance scales as a^2: $Z_1 = a^2 Z_2$. In audio systems, transformers match unequal impedances to maximize transferred power.

CH07-24 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Formula Sheet

Transformer: Open-Circuit (No-Load) Test for Shunt Branch Parameters

PROBLEM / QUESTION:

In a transformer Open-Circuit Test: (1) Which winding is energized and which is left open? (2) Which equivalent circuit parameters are determined, and what does the wattmeter reading represent?

KEY FORMULAS & DEFINITIONS:

$$P_{oc} \approx P_{\text{core}}, \quad R_c = \frac{V_{oc}^2}{P_{oc}}$$

$$I_c = \frac{V_{oc}}{R_c}, \quad I_m = \sqrt{I_{oc}^2 - I_c^2}, \quad X_m = \frac{V_{oc}}{I_m}$$

STEP-BY-STEP MODEL SOLUTION:

1. Connection:
- Low Voltage (LV) side is energized at rated voltage.
- High Voltage (HV) side is left completely OPEN for operator safety.

2. Measured Values & Parameters:
- Wattmeter measures Core Loss directly ($P_{oc} = P_{\text{core}}$) because no-load current is tiny ($< 3-5\%$), making winding I^2R negligible.
- Yields shunt excitation branch parameters: Core resistance R_c (modeling hysteresis & eddy losses) and magnetizing reactance X_m (modeling core magnetization).

EXAM PITFALL & CAUTION:

Open-Circuit test yields CORE LOSS (P_{core}) and shunt branch (R_c, X_m). Do NOT confuse with short-circuit test!

CH07-25 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Formula Sheet

Transformer: Short-Circuit Test for Series Winding Parameters

PROBLEM / QUESTION:

In a transformer Short-Circuit Test: (1) Which winding is short-circuited and which is energized? (2) Which equivalent circuit parameters are determined, and what does the wattmeter reading represent?

KEY FORMULAS & DEFINITIONS:

$$P_{sc} \approx P_{\text{cu}}, \quad R_{eq} = \frac{P_{sc}}{I_{sc}^2}$$

$$Z_{eq} = \frac{V_{sc}}{I_{sc}}, \quad X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

STEP-BY-STEP MODEL SOLUTION:

1. Connection:
- Low Voltage (LV) side is securely SHORT-CIRCUITED with a thick jumper bar.
- High Voltage (HV) side is energized with a REDUCED voltage ($5-10\%$ of rated) adjusted until rated current flows.

2. Measured Values & Parameters:
- Wattmeter measures Full-Load Copper Loss ($P_{sc} = P_{\text{cu}}$) because applied voltage is so low that core flux and core loss are negligible.
- Yields series equivalent parameters: Equivalent resistance $R_{eq} = R_1 + a^2 R_2$ and equivalent leakage reactance $X_{eq} = X_1 + a^2 X_2$.

EXAM PITFALL & CAUTION:

Short-Circuit test yields COPPER LOSS (P_{cu}) and series parameters (R_{eq}, X_{eq}). Never apply full rated voltage during a short-circuit test!

CH07-26 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Transformer Voltage Regulation Formula & Power Factor Dependency

📌 **PROBLEM / QUESTION:**

State the formula for Transformer Voltage Regulation ($VR\%$). How does the power factor (lagging vs leading) affect voltage regulation, and can $VR\%$ ever be negative?

📐 **KEY FORMULAS & DEFINITIONS:**

$$VR\% = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

Approx:

$$VR\% \approx \frac{I_2 R_{eq2} \cos\theta \pm I_2 X_{eq2} \sin\theta}{V_2} \times 100\%$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Formula:

$$VR\% = \frac{V_{2,NL} - V_{2,FL}}{V_{2,FL}} \times 100\%$$

2. Effect of Power Factor:
- Lagging PF (Inductive load): (+) sign applies. Voltage drops under load ($V_{FL} < V_{NL}$), so $VR\% > 0$ (positive).
- Unity PF ($PF = 1.0$): Small positive $VR\%$.
- Leading PF (Capacitive load): (-) sign applies. Due to capacitive Ferranti effect, terminal voltage rises under load ($V_{FL} > V_{NL}$), causing $VR\% < 0$ (NEGATIVE)!

⚠️ **EXAM PITFALL & CAUTION:**

Negative voltage regulation ($VR\% < 0$) happens ONLY with LEADING power factors (capacitive loads), meaning voltage at full load is HIGHER than at no load!

CH07-27 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Final Exam

Auto-Transformer Principle, Higher kVA Rating & Common Winding Risk

PROBLEM / QUESTION:

What is an Auto-Transformer? Why does it possess a higher kVA power rating and higher efficiency than a conventional two-winding transformer of the same physical size, and what is its primary safety drawback?

KEY FORMULAS & DEFINITIONS:

$$\frac{S_{\text{auto}}}{S_{\text{2-w}}} = \frac{1}{1 - 1/a_{\text{auto}}}$$

$$S_{\text{auto}} = S_{\text{conducted}} + S_{\text{transformed}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Construction & Principle:
An auto-transformer has a SINGLE continuous winding shared partly by both primary and secondary circuits (tapped common winding).

2. Higher Rating & Efficiency:
Power is transferred via two mechanisms simultaneously:
- By direct electrical conduction ($S_{\text{conducted}} = V_2 I_1$)
- By electromagnetic induction ($S_{\text{transformed}} = V_2(I_2 - I_1)$)
Because a large fraction of power flows conductively without transforming through magnetic flux:

$$\frac{S_{\text{auto}}}{S_{\text{2-w}}} = \frac{1}{1 - 1/a} > 1$$

Core and windings are much smaller, dramatically lowering I^2R heat loss and core losses.

3. Safety Drawback:
NO ELECTRICAL ISOLATION between primary (HV) and secondary (LV)! If the common ground connection opens, the full high voltage appears directly on the low voltage terminals, risking electrocution.

EXAM PITFALL & CAUTION:

Never use an autotransformer where human safety isolation from high voltage mains is required (e.g. medical equipment or bathroom supplies).

PART 3

Chapter 8: Power Distribution System

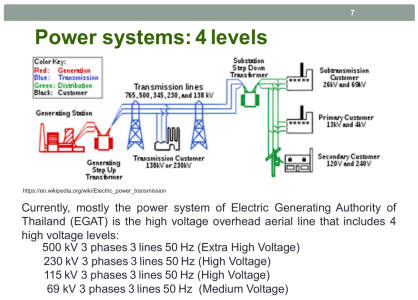
EGAT 4 Voltage Levels, EGAT 5 Power Plant Types, MEA & PEA Networks, Service Connections & Substations (15 Flashcards)

CH08-01 | Chapter 8 | Level 1 (Core Concept) | 📌 Ch08 Slide 2, 7, 8 & Final Exam ข้อ 5.3 (หน้า 25)

Structure of Electrical Power Systems: 4 Stages & EGAT Voltage Levels

📌 PROBLEM / QUESTION:

State the 4 main stages of an electrical power system from generation to consumers. Explain why generators generate power at 10-20 kV, why voltage must be stepped up for transmission, and list EGAT's 4 standard transmission voltage levels.



รูปประกอบ: Slide 7: EGAT 4 High-Voltage Transmission Levels & Power Systems Structure

📐 KEY FORMULAS & DEFINITIONS:

$$\text{Stages: Generation (10-20 kV)} \longrightarrow \text{Transmission (69-500 kV)} \longrightarrow \text{Distribution (12-33 kV)} \longrightarrow \text{Utilization (230/400 V)}$$

$$\text{EGAT Transmission Levels: 500 kV (EHV),} \quad \text{230 kV (HV),} \quad \text{115 kV (HV),} \quad \text{69 kV (MV/HV)}$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Four Main Stages of Electrical Power System:
- Generation System (ระบบการผลิต): Generates electricity at 10-20 kV. Limited to this range because of generator winding insulation limitations.
- Transmission System (ระบบสายส่ง): Steps up voltage to Extra High Voltage (EHV) and High Voltage (HV) using step-up transformers to minimize line current, I^2R transmission losses, and voltage drop over long distances.
- Distribution System (ระบบจำหน่าย): Substations step down high voltage to Medium Voltage (MV) (24/12 kV for MEA, 22/33 kV for PEA) for urban and regional distribution.
- Utilization System (ระบบการใช้ไฟฟ้า): Distribution transformers step down MV to Low Voltage (LV) (3-phase 400/230 V, 1-phase 230 V) for consumers.

2. EGAT's 4 High-Voltage Overhead Transmission Levels:
- 500 kV, 3-phase 3-wire, 50 Hz (Extra High Voltage: EHV)
- 230 kV, 3-phase 3-wire, 50 Hz (High Voltage: HV)
- 115 kV, 3-phase 3-wire, 50 Hz (High Voltage: HV)
- 69 kV, 3-phase 3-wire, 50 Hz (Medium Voltage / High Voltage)

⚠️ EXAM PITFALL & CAUTION:

Generation voltage is strictly 10-20 kV due to generator insulation. Transmission step-up is mandatory before sending power over long distances to keep I^2R losses low.

CH08-02 | Chapter 8 | Level 2 (Exam / Calc) | Ch08 Slide 8, 38 & Final Exam Cheatsheet

Power Transmission Loss Formula & High-Voltage Economics

PROBLEM / QUESTION:

Derive the mathematical relationship between transmission line voltage V_L, line current I_L, and I^2R power losses for transmitting real power P. Explain quantitatively why stepping up voltage by 10 times dramatically reduces conductor losses and cross-sectional material costs.

KEY FORMULAS & DEFINITIONS:

$$I_L = \frac{P}{\sqrt{3}\, V_L \cos\theta}$$

$$P_{\text{loss}} = 3I_L^2 R_{\text{line}} = 3\left(\frac{P}{\sqrt{3}\, V_L \cos\theta}\right)^2 R_{\text{line}} = \frac{P^2 R_{\text{line}}}{V_L^2 \cos^2\theta}$$

$$P_{\text{loss}} \propto \frac{1}{V_L^2}, \qquad \Delta V = \sqrt{3}\, I_L (R\cos\theta + X\sin\theta) \propto \frac{1}{V_L}$$

STEP-BY-STEP MODEL SOLUTION:

1. Derivation of Line Current:

For a balanced 3-phase line transmitting active power P at power factor $\cos\theta$:

$$P = \sqrt{3}\, V_L I_L \cos\theta \implies I_L = \frac{P}{\sqrt{3}\, V_L \cos\theta}$$

2. Line Losses (P_{loss}):

Total Joule heating loss across all 3 phases:

$$P_{\text{loss}} = 3I_L^2 R_{\text{line}} = \frac{P^2 R_{\text{line}}}{V_L^2 \cos^2\theta}$$

3. Quantitative Impact of 10x Voltage Step-Up:

- Line current I_L decreases to $\frac{1}{10}$ (10 times smaller).
- Line power loss P_{loss} decreases by $10^2 = 100$ times (99% reduction in loss!).
- Conductor cross-sectional area A required to carry I_L decreases proportionally, drastically cutting copper/aluminum conductor mass and tower structural weight.

EXAM PITFALL & CAUTION:

Transmission losses depend inversely on the SQUARE of line voltage ($1/V_L^2$), while voltage drop ΔV is inversely proportional to V_L.

CH08-03 | Chapter 8 | Level 1 (Core Concept) | 📌 Ch08 Slide 5, 6, 10, 11 & Final Exam ข้อ 5.1

Thailand Three Electricity Authorities & National Consumption Share

📌 **PROBLEM / QUESTION:**

Name the 3 state electricity authorities of Thailand, specify their jurisdictional boundaries and core roles, and present Thailand's national electricity consumption share and primary generation fuel mix.

📐 **KEY FORMULAS & DEFINITIONS:**

Consumption Share: PEA (กฟภ.) 72%, MEA (กฟน.) 27%, EGAT Direct Customers 1%

Fuel Mix: Natural Gas 57%, Coal/Lignite 18%, Import 12%, Renewable 9%, Hydro 4%

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Three State Electricity Authorities:
- EGAT (Electricity Generating Authority of Thailand / กฟผ.): Responsible for national bulk power generation and nationwide high-voltage transmission grid (500, 230, 115, 69 kV). Sells wholesale electricity to MEA and PEA; manages international power import/export with Laos, Malaysia, etc.
- MEA (Metropolitan Electricity Authority / กฟน.): Distributes electricity strictly within 3 metropolitan provinces: Bangkok, Nonthaburi, and Samut Prakan.
- PEA (Provincial Electricity Authority / กฟภ.): Distributes electricity to all remaining 74 provinces across Thailand.

2. Electricity Consumption Share (2561 Data):
- PEA: 72% (77,953 GWh)
- MEA: 27% (29,935 GWh)
- Direct EGAT Industrial Customers: 1% (874 GWh)

3. Primary Fuel Mix for Power Generation in Thailand:
- Natural Gas: 57% (Primary fuel)
- Imported Coal & Lignite: 18%
- Imported Electricity (e.g. hydro from Laos): 12%
- Renewable Energy (Solar, Wind, Biomass): 9%
- Domestic Hydro: 4%
- Petroleum Oil: 0.1%

⚠️ **EXAM PITFALL & CAUTION:**

MEA covers ONLY 3 provinces (Bangkok, Nonthaburi, Samut Prakan). All other 74 provinces belong to PEA. Natural Gas is Thailand's primary fuel (57%).

CH08-04 | Chapter 8 | Level 2 (Exam / Calc) | 📌 Ch08 Slide 13, 21, 24 & Final Exam ข้อ 5.2 (หน้า 25)

Base Load vs Peak Load Power Plants: Comparison & Startup Times

📌 **PROBLEM / QUESTION:**

Compare Base Load and Peak Load power plants regarding operating objectives, fuel types, capital vs fuel costs, and startup times, citing typical Thai power plant examples.

📐 **KEY FORMULAS & DEFINITIONS:**

Base Load: 24/7 continuous, High Capital Cost, Low Fuel Cost/kWh, Startup: 6-8 Hours (Mae Moh Lignite)

Peak Load: Fast response, Low Capital Cost, High Fuel Cost/kWh, Startup: $<$ 5-15 Minutes (Hydro, Gas Turbine)

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Base Load Power Plants (โรงไฟฟ้าฐาน):
- Operating Objective: Run continuously 24 hours a day at constant rated output to cover minimum baseline electrical demand.
- Economic Profile: High capital / construction cost, but low fuel cost per generated kWh.
- Startup Time: Very slow (6 to 8 hours) due to thermal boiler expansion requirements.
- Fuels & Examples: Lignite coal (Mae Moh Power Plant, Lampang), Nuclear, Combined-Cycle baseload.

2. Peak Load Power Plants (โรงไฟฟ้ารับโหลดสูงสุด):
- Operating Objective: Operated only during hours of peak demand (18:00 - 22:00) or emergency power deficits.
- Economic Profile: Low capital construction cost, but high fuel / operating cost per kWh.
- Startup Time: Extremely rapid (under 5 minutes for Hydro; ~15 minutes for Gas Turbine).
- Fuels & Examples: Hydroelectric dam (Bhumibol, Ubolratana), Pumped-Storage Hydro (Lam Takhong), Open-Cycle Gas Turbine, Diesel.

⚠️ **EXAM PITFALL & CAUTION:**

Hydroelectric dams start in under 5 minutes and are ideal for Peak Load. Steam coal plants require 6-8 hours and can only serve as Base Load.

CH08-05 | Chapter 8 | Level 1 (Core Concept) | 📌 Ch08 Slide 12-34 & Final Exam ข้อ 5.1

Five Types of EGAT Power Plants & Landmark Facilities in Thailand

📌 **PROBLEM / QUESTION:**

Name the 5 types of electrical power plants operated by EGAT, and describe the key landmark facilities in Thailand (Bhumibol Dam, Ubolratana Dam, Mae Moh, Nam Pong, and Bangpakong).

📐 **KEY FORMULAS & DEFINITIONS:**

1. Hydro (Bhumibol 7,312 MW, Ubolratana 2,263M m^3, Lam Takhong Pumped Storage)
2. Steam/Thermal (Mae Moh 2,625 MW), 3. Gas/Diesel (Nam Pong 710 MW)
4. Combined Cycle (Bangpakong Pipeline 8 km), 5. Renewable (Solar, Wind, Biomass)

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Five EGAT Power Plant Types:
- 1) Hydro Power Plant (โรงไฟฟ้าพลังน้ำ): Rapid start (< 5 min), stores water in dam. Landmark dams:
- Bhumibol Dam (Tak): Thailand's first multi-purpose dam (formerly Yanhee Dam), only curved concrete arch dam in Thailand (radius 250 m, height 154 m, reservoir 13,462 million m^3, 8 generators, 7,312 MW).
- Ubolratana Dam (Khon Kaen): Built across Pong River, first hydroelectric dam in Northeastern Thailand (crest length 885 m, height 32 m, reservoir 2,263 million m^3).
- Lam Takhong Power Plant: Underground pumped-storage hydropower plant.
- 2) Steam / Thermal Power Plant (โรงไฟฟ้าพลังความร้อน/ไอน้ำ): High capacity, 6-8 hr startup, base load. Landmark: Mae Moh Power Plant (Lampang) with 13 generators, 2,625 MW capacity using local lignite coal.
- 3) Gas and Diesel Power Plant (โรงไฟฟ้ากังหันก๊าซและดีเซล): 15 min startup, peak load. Landmark: Nam Pong Power Plant (Khon Kaen, 710 MW using natural gas, 32 km from KKU downtown).
- 4) Combined Cycle Power Plant (โรงไฟฟ้าพลังความร้อนร่วม): Combines Gas Turbine + Steam Turbine with Heat Recovery Steam Generator (HRSG). Landmark: Bangpakong Power Plant (supplied by 8 km heavy oil pipeline from Sriracha depot).
- 5) Renewable Power Plant (โรงไฟฟ้าพลังงานหมุนเวียน): Solar, wind, biomass.

⚠️ **EXAM PITFALL & CAUTION:**

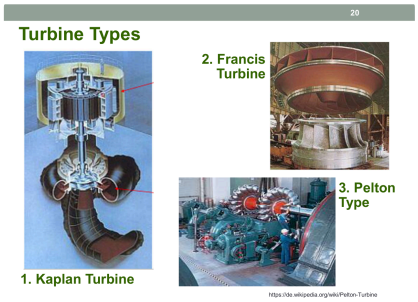
Bhumibol Dam is Thailand's ONLY curved concrete arch dam. Ubolratana Dam is the FIRST hydro dam in the Northeast (Isan), located right in Khon Kaen.

CH08-06 | Chapter 8 | Level 1 (Core Concept) | 📌 Ch08 Slide 20 (Turbine Types)

Hydraulic Turbine Classifications: Pelton, Francis & Kaplan

📌 PROBLEM / QUESTION:

Classify the 3 major types of hydraulic turbines (Pelton, Francis, Kaplan) by working principle, water head (H), flow rate (Q), and their engineering applications in hydroelectric dams.



รูปประกอบ: Slide 20: Three Hydraulic Turbine Types (Kaplan, Francis, Pelton)

📐 KEY FORMULAS & DEFINITIONS:

$$\text{Pelton (Impulse): High Head } (H > 300 \text{ m}), \quad \text{Low Flow Rate } (Q)$$
$$\text{Francis (Reaction): Medium Head } (30 \text{ m} \leq H \leq 300 \text{ m}), \quad \text{Medium Flow Rate } (Q)$$
$$\text{Kaplan (Propeller): Low Head } (H < 30 \text{ m}), \quad \text{High Flow Rate } (Q)$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Pelton Turbine (Impulse Turbine / กังหันแรงกระแทก):
- Operating Principle: High-velocity water jets from nozzles strike cup-shaped buckets on the runner periphery in open air.
- Operating Conditions: High Head ($H > 300$ m) and Low Flow Rate (Q).
- Typical Application: High mountainous dams with steep waterfalls.

2. Francis Turbine (Reaction Turbine / กังหันแรงสะท้อน):
- Operating Principle: Water enters the runner radially under pressure through wicket gates and discharges axially.
- Operating Conditions: Medium Head ($30 \text{ m} \leq H \leq 300$ m) and Medium Flow Rate (Q).
- Typical Application: Most widely used turbine globally, applied in large multi-purpose dams (e.g. Bhumibol Dam, Srinagarind Dam).

3. Kaplan Turbine (Propeller Reaction Turbine / กังหันแบบใบพัด):
- Operating Principle: Axial-flow propeller runner with adjustable blades (pitch control) to maintain high efficiency over variable flow rates.
- Operating Conditions: Low Head ($H < 30$ m) and High Flow Rate (Q).
- Typical Application: River run-of-river dams and low-head barrages (e.g. Chao Phraya Dam).

⚠️ EXAM PITFALL & CAUTION:

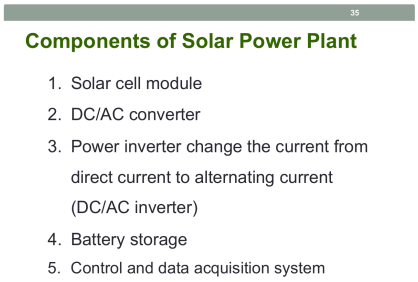
Pelton = High Head, Low Flow. Francis = Medium Head/Flow (most common). Kaplan = Low Head, High Flow (adjustable propeller).

CH08-07 | Chapter 8 | Level 1 (Core Concept) | 📌 Ch08 Slide 33, 35, 36 (Solar PV System)

Solar Photovoltaic (PV) Power System Core Components

📌 **PROBLEM / QUESTION:**

List the 5 primary components of a complete solar photovoltaic power plant, explain the role of each, and name notable solar demonstration plants in Thailand.



รูปประกอบ: Slide 35: Components of a Solar Power Plant

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{1. Solar PV Modules (DC)} \longrightarrow \text{2. DC/AC Inverter} \longrightarrow \text{AC Grid / Load (230/400 V)}$$

$$\text{Auxiliary: 3. Battery Storage (BESS),} \quad \text{4. Charge Controller,} \quad \text{5. SCADA / Data Acquisition}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Five Primary Components of Solar PV System:
- 1) Solar Cell Modules (แผงโซลาร์เซลล์): Converts incident sunlight directly into direct current (DC) electricity via the semiconductor photovoltaic effect.
- 2) DC/AC Inverter (อินเวอร์เตอร์): Converts generated DC power into synchronized Alternating Current (AC) power at 50 Hz, 230/400 V matching grid standards.
- 3) Battery Energy Storage (ระบบแบตเตอรี่กักเก็บพลังงาน): Stores excess generated electrical energy during daytime peak irradiance for nighttime or emergency power supply.
- 4) Charge Controller & Protection Switchgear: Regulates battery charging/discharging voltage and current to prevent overcharging and damage.
- 5) Control and Data Acquisition System (SCADA / ระบบตรวจวัด): Monitored parameters include solar irradiance, cell temperatures, DC/AC voltages, power output, and inverter conversion efficiency.

2. Notable Solar PV Plants in Thailand (Ch08 Slide 33):
- Klongchongklam (คลองช่องกล่ำ จ.สระแก้ว): 20 kW demonstration plant
- Promthep Cape (แหลมพรหมเทพ จ.ภูเก็ต): 11.34 kW
- Sankamphaeng (สันกำแพง จ.เชียงใหม่): 14 kW
- Mae Hong Son (จ.แม่ฮ่องสอน): 500 kW

⚠️ **EXAM PITFALL & CAUTION:**

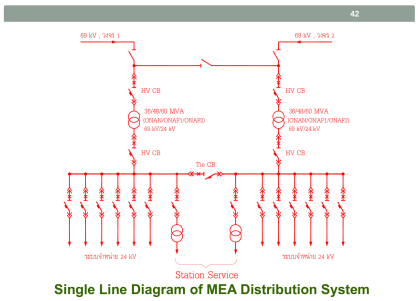
Inverters are DC-to-AC converters. The 5 components specified in slide 35 are: PV Module, DC/AC Inverter, Battery Storage, Charge Controller, and SCADA System.

CH08-08 | Chapter 8 | Level 2 (Exam / Calc) | Ch08 Slide 41, 42, 43 (MEA Distribution System)

MEA Distribution System Architecture & Transformer Ratings

PROBLEM / QUESTION:

Detail the distribution system of the Metropolitan Electricity Authority (MEA), including service provinces, voltage levels, substation redundancy, power transformer capacities, feeder load limits, and secondary utilization voltages.



รูปประกอบ: Slide 42: Single Line Diagram of MEA Distribution System (69 kV / 24 kV)

KEY FORMULAS & DEFINITIONS:

$$\text{Subtransmission: 69 or 115 kV} \xrightarrow{\text{Substation}} \text{Primary Distribution: 24 kV (New) or 12 kV (Legacy)}$$

$$\text{Substation Transformers: } 36/48/60 \text{ MVA (ONAN/ONAF1/ONAF2) with Tie CB}$$

$$\text{Feeder Capacity: } 15 \text{ MVA (at 24 kV)}, \quad 8 \text{ MVA (at 12 kV)}$$

$$\text{Utilization: } 24/12 \text{ kV} \longrightarrow 416/240 \text{ V rated} \implies \text{Delivered: } 380/220 \text{ V or } 400/230 \text{ V (3-phase 4-wire)}$$

STEP-BY-STEP MODEL SOLUTION:

1. MEA Jurisdiction & Subtransmission:
- Covers 3 provinces: Bangkok, Samut Prakan, and Nonthaburi.
- Sub-transmission System: Receives bulk power from EGAT at 230, 115, or 69 kV.

2. MEA Distribution Substations:
- Stepping Down: Transforms 69 kV or 115 kV down to 24 kV (standard current system) or 12 kV (older legacy system).
- Circuit Redundancy: Powered via 2 independent incoming circuits for continuous reliability.
- Transformer Ratings: Uses 2 to 4 power transformers rated at 36/48/60 MVA (ONAN/ONAF1/ONAF2) per circuit, linked by a High-Voltage Tie Circuit Breaker (Tie CB).
- Feeder Load Capacities: Maximum 8 MVA per feeder at 12 kV; maximum 15 MVA per feeder at 24 kV.

3. MEA Utilization System (Low Voltage Delivery):
- Distribution transformers step down 24 kV (or 12 kV) to secondary rated voltage of 416/240 V.
- Standard voltage delivered to end consumers is 380/220 V (or 400/230 V), 3-phase 4-wire with neutral.

EXAM PITFALL & CAUTION:

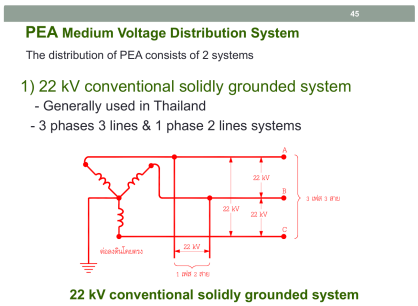
MEA uses 24 kV (new) and 12 kV (legacy) for medium voltage. Transformer rating is 36/48/60 MVA with Tie CB. Secondary rating is 416/240 V delivering 380/220 V.

CH08-09 | Chapter 8 | Level 2 (Exam / Calc) | Ch08 Slide 45, 46 (PEA Medium Voltage Systems)

PEA Medium Voltage Distribution: 22 kV Solidly Grounded vs 33 kV Multigrounded

PROBLEM / QUESTION:

Compare the two distinct Medium Voltage distribution systems used by PEA (22 kV Solidly Grounded vs 33 kV Multigrounded with Overhead Ground Wire) in terms of geographic coverage, line configuration, phase-to-ground voltage, and earthing method.



รูปประกอบ: Slide 45 & 46: PEA 22 kV Solidly Grounded vs 33 kV Multigrounded System

KEY FORMULAS & DEFINITIONS:

$$\text{System 1: 22 kV Conventional Solidly Grounded} \implies \text{3-Phase 3-Wire, } V_L = 22 \text{ kV (Central, North, Northeast, East)}$$

$$\text{System 2: 33 kV Multigrounded with OHGW} \implies \text{3-Phase 4-Wire, } V_L = 33 \text{ kV}, \quad V_p = \frac{33}{\sqrt{3}} \approx 19 \text{ kV}$$

\text{(Used in South: Ranong southward, plus Chiang Rai & Phayao; grounded at every pole)}

STEP-BY-STEP MODEL SOLUTION:

1. PEA 22 kV Conventional Solidly Grounded System (สไลด์ที่ 45):
- Geographic Coverage: Standard system applied throughout Central, Northern, Northeastern (Isan), and Eastern regions of Thailand.
- Line Configuration: 3-phase 3-wire system ($V_L = 22$ kV) and 1-phase 2-wire branch lines.
- Earthing Method: Solidly grounded at the substation transformer star neutral point.

2. PEA 33 kV Multigrounded System with Overhead Ground Wire (สไลด์ที่ 46):
- Geographic Coverage: Applied specifically in Southern Thailand (from Ranong province southward) and upper Northern border provinces (Chiang Rai, Phayao).
- Line Configuration: 3-phase 4-wire system with Line-to-Line voltage $V_L = 33$ kV and Phase-to-Ground voltage $V_p = \frac{33}{\sqrt{3}} \approx 19$ kV.
- Earthing & Ground Wire: The 4th overhead conductor serves as an Overhead Ground Wire (OHGW) / neutral, grounded at the substation star point and *grounded at every single utility pole*. This provides superior shielding against lightning strikes in high thunder-day coastal regions.

EXAM PITFALL & CAUTION:

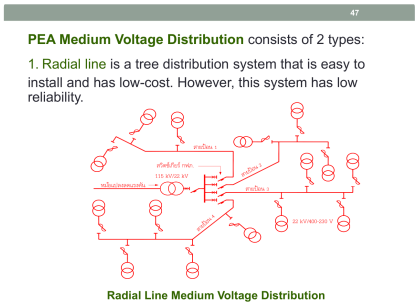
PEA uses 22 kV (3-phase 3-wire) for most of Thailand, but uses 33 kV (3-phase 4-wire with OHGW grounded at every pole) in the South (Ranong down) and Chiang Rai/Phayao.

CH08-10 | Chapter 8 | Level 2 (Exam / Calc) | Ch08 Slide 47, 48 (PEA Feeder Topologies)

PEA Distribution Feeder Topologies: Radial vs Tie-Line Systems

PROBLEM / QUESTION:

Contrast the Radial Line and Tie-Line feeder configurations used in PEA medium voltage networks, evaluating their construction complexity, capital cost, and reliability during feeder faults.



รูปประกอบ: Slide 47 & 48: Radial Line vs Tie-Line Distribution Topologies

KEY FORMULAS & DEFINITIONS:

$$\text{Radial Line: Simple Tree} \longrightarrow \text{Lowest Cost, Easy Protection, } \mathbf{\text{Low Reliability}}$$

$$\text{Tie Line: Interconnected Feeders with Tie CB} \longrightarrow \text{Higher Cost, Flexible Back-feed, } \mathbf{\text{High Reliability}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Radial Line Medium Voltage Distribution (ระบบสายป้อนแบบรัศมี / กิ่งไม้ - สไลด์ที่ 47):
- Topology: Single feeder originates from the substation switchgear and branches outwards like tree branches to supply distribution transformers.
- Advantages: Simplest layout, lowest construction cost, straightforward protection coordination (unidirectional fault current).
- Disadvantages: Low reliability. Any short circuit or line break on the trunk feeder causes the upstream breaker to trip, interrupting power to all downstream customers without alternative supply path.

2. Tie Line Medium Voltage Distribution (ระบบสายป้อนแบบเชื่อมโยงระหว่างสายป้อน - สไลด์ที่ 48):
- Topology: Two or more feeders originating from different substations (Substation A and Substation B) or different busbars are connected via normally-open Tie Switches or Tie Circuit Breakers.
- Advantages: High reliability. When a fault occurs on feeder A, the faulted section can be isolated via sectionalizing switches, and the remaining healthy portion can be back-fed by closing the Tie Switch to Substation B.
- Disadvantages: Higher capital cost for tie lines, reclosers, automated switchgear, and complex bidirectional protection schemes.

EXAM PITFALL & CAUTION:

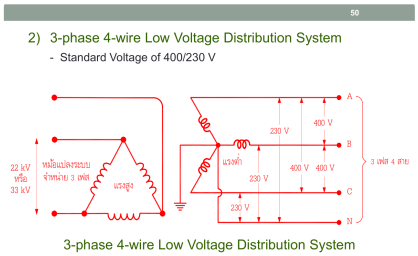
Radial = Low cost, Low reliability (tree). Tie Line = Higher cost, High reliability (can switch power from adjacent substation during outages).

CH08-11 | Chapter 8 | Level 1 (Core Concept) | 📌 Ch08 Slide 49, 50 (PEA Low Voltage Distribution)

PEA Low Voltage Distribution Systems (1-Phase vs 3-Phase Standard)

📌 PROBLEM / QUESTION:

Describe PEA's Low Voltage distribution systems, explaining the configurations and specific use cases of 1-phase 2-wire, 1-phase 3-wire, and 3-phase 4-wire systems.



รูปประกอบ: Slide 50: PEA 3-Phase 4-Wire Low Voltage Distribution System (400/230 V)

📐 KEY FORMULAS & DEFINITIONS:

$$\text{1-Phase 2-Wire: 230 V (Standard Residential)}$$
$$\text{1-Phase 3-Wire: 460/230 V (Center-tapped, Street Lighting)}$$
$$\text{3-Phase 4-Wire: 400/230 V } (V_L = 400 \text{ V}, \quad V_p = 230 \text{ V, Commercial \& Motor Loads})$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Single-Phase Low Voltage Systems (Ch08 Slide 49):
- 1-Phase 2-Wire (230 V): Standard residential service. One phase wire and one neutral wire at 230 V, 50 Hz.
- 1-Phase 3-Wire (460/230 V): Supplied by center-tapped single-phase distribution transformer. Phase-to-neutral is 230 V, while phase-to-phase across the outer legs is 460 V. Widely used for street lighting circuits to transmit power twice as far with acceptable voltage drop.

2. Three-Phase Low Voltage System (Ch08 Slide 50):
- 3-Phase 4-Wire (400/230 V standard): Secondary side of 22 kV or 33 kV distribution transformers connected in Wye (Star) with grounded neutral (N).
- Line-to-Line Voltage: $V_L = 400$ V
- Phase-to-Neutral Voltage: $V_p = \frac{400}{\sqrt{3}} \approx 230$ V
- Supplies 3-phase commercial facilities, workshops, water pumps, industrial induction motors, and balanced 3-phase distribution networks.

⚠️ EXAM PITFALL & CAUTION:

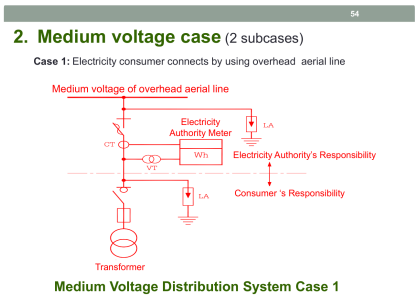
1-phase 3-wire 460/230 V is specially used for street lighting circuits. Standard 3-phase low voltage in Thailand is 400/230 V (IEC standard).

CH08-12 | Chapter 8 | Level 2 (Exam / Calc) | Ch08 Slide 51-55 (Service Connection & Metering)

Electricity Service Connection, Demand Levels & Standard Meter Sizes

PROBLEM / QUESTION:

State the electricity service connection rules based on customer power demand, list standard meter ratings in Thailand, and define the point of delivery / boundary of responsibility between the utility and the consumer for Low Voltage vs Medium Voltage connections.



รูปประกอบ: Slide 54: Medium Voltage Distribution Connection & Boundary of Responsibility

KEY FORMULAS & DEFINITIONS:

$$\text{Load} < 300\ \text{kVA} \longrightarrow \text{Low Voltage } (230/400\ \text{V})$$
$$\text{Load} > 300\ \text{kVA} \longrightarrow \text{Medium Voltage } (12, 22, 24, \text{or } 33\ \text{kV with CT/VT})$$
$$\text{Very Large Establishment} \longrightarrow \text{High Voltage } (115\ \text{kV with Customer Dedicated Substation})$$
$$\text{Meter Ratings (1-Phase 220V): } 5(15)\ \text{A}, \quad 15(45)\ \text{A}, \quad 30(100)\ \text{A}, \quad 50(150)\ \text{A}$$
$$\text{Meter Ratings (3-Phase 380V): } 15(45)\ \text{A}, \quad 30(100)\ \text{A}, \quad 50(150)\ \text{A}, \quad 200\ \text{A (CT)}, \quad 400\ \text{A (CT)}$$

STEP-BY-STEP MODEL SOLUTION:

1. Customer Connection Voltage based on Load Demand (Slide 51):
- Load < 300 kVA: Supplied at Low Voltage (230/400 V) from utility pole distribution transformers.
- Large Establishment (> 300 kVA): Supplied at Medium Voltage (12 kV, 22 kV, 24 kV, or 33 kV). Customer must install dedicated transformer.
- Very Large Establishment: Supplied at High Voltage (115 kV). Customer must construct their own private substation.

2. Standard Meter Sizes in Thailand (Slide 52):
- 1-Phase 220 V: 5(15) A, 10(30) A, 15(45) A, 30(100) A, 50(150) A. (Number in parentheses is peak thermal withstand current).
- 3-Phase 380/220 V: 15(45) A, 30(100) A, 50(150) A; loads above 150 A require Current Transformers (CT) with 200 A or 400 A meters.
- MV 12 kV: 15 A (300 kVA) up to 750 A (15,000 kVA).
- MV 24 kV: 10 A (400 kVA) up to 625 A (25,000 kVA).

3. Points of Delivery & Boundaries of Responsibility (Slides 53-55):
- Low Voltage: Utility owns and maintains lines up to the kWh meter on the pole. Customer is responsible for service entrance wires, main switch / MDB, and internal wiring.
- Medium Voltage Case 1 (Overhead Line Connection): Utility installs Lightning Arresters (LA), CT, VT, and primary kWh meter. Utility responsibility ends at meter pole; customer owns and maintains transformer, drop-out fuses, and secondary distribution.
- Medium Voltage Case 2 (Underground Cable Connection): High-voltage switchgear and underground cables transition power to customer transformer; utility metering is done via MV CT/VT.

EXAM PITFALL & CAUTION:

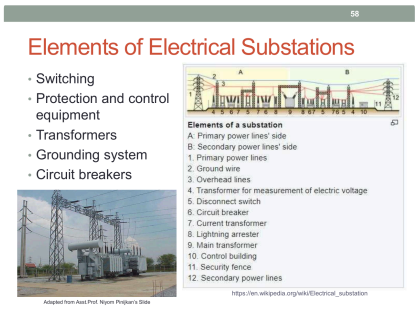
Load boundary: < 300 kVA is Low Voltage; > 300 kVA is Medium Voltage. High Voltage requires a private substation (115 kV).

CH08-13 | Chapter 8 | Level 1 (Core Concept) | 📌 Ch08 Slide 56-59 (Electrical Substations)

Electrical Substations: Core Functions, Elements & AIS vs GIS Types

📌 PROBLEM / QUESTION:

Explain the 4 primary functions of an electrical substation, list its core equipment, and compare Air-Insulated Substations (AIS / Outdoor) with Gas-Insulated Substations (GIS / Indoor, e.g. KKU Substation).



รูปประกอบ: Slide 58 & 59: Elements of Substations and KKU Indoor Substation

📐 KEY FORMULAS & DEFINITIONS:

Functions: 1. Voltage Transformation, 2. Switching, 3. Fault Clearance/Protection, 4. Voltage/VAR Control

Core Elements: Power Transformers, Circuit Breakers (SF_6/Vacuum), Disconnectors, CT/VT, Relays, LA, Grounding Grid

AIS (Outdoor): Air insulation, large footprint, lower initial cost, weather-exposed

GIS (Indoor): SF_6 gas (high dielectric strength), 85-90% smaller footprint, enclosed, silent (e.g. KKU Substation)

📝 STEP-BY-STEP MODEL SOLUTION:

1. Four Primary Functions of Electrical Substations (Ch08 Slide 57):
- 1) Voltage Transformation: Steps up voltage from generators to transmission grid or steps down from transmission to medium-voltage distribution circuits.
- 2) Switching Operations: Connects and disconnects transmission lines, busbars, and feeders for power routing and load balancing.
- 3) System Protection: Automatically clears short-circuit faults via circuit breakers and protective relays to safeguard equipment.
- 4) Voltage Regulation and Reactive Power Support: Utilizes shunt capacitor banks and shunt reactors to maintain stable grid voltage.

2. Seven Substation Classifications:
Transmission substation, Distribution substation, Collector substation (for wind/solar), Converter substation (HVDC AC-to-DC), Switching station, Traction substation (electric rail/trains), Mobile substation.

3. Core Equipment in a Substation (Slide 58):
- Power Transformers (หม้อแปลงกำลัง)
- Circuit Breakers (สวิตช์ตัดตอนอัตโนมัติ: SF_6, vacuum, oil, air)
- Disconnecting Switches / Isolators (ใบมีดตัดตอน)
- Instrument Transformers (Current Transformers: CT, Potential/Voltage Transformers: PT/VT)
- Protective Relays & SCADA Control Panels
- Lightning Arresters (กับดักเสิร์จ/ฟ้าผ่า)
- Grounding Grid (ตะแกรงระบบต่อลงดิน)

4. Outdoor (AIS) vs Indoor (GIS) Substations (Slide 59):
- Air-Insulated Substation (AIS / Outdoor): Uses atmospheric air as insulating medium. Requires large clearance distances, resulting in a large land footprint. Cheaper equipment, but exposed to weather, lightning, and atmospheric pollution.
- Gas-Insulated Substation (GIS / Indoor): Encapsulates all conductors and switchgear inside grounded metal enclosures filled with Sulfur Hexafluoride (SF_6) gas. SF_6 has 2.5-3 times higher dielectric strength than air. Reduces

land area requirements by 85-90%. Silent, safe, fireproof, protected from environment, ideal for densely populated urban centers and university campuses (e.g. KKU Indoor Substation).

⚠️ **EXAM PITFALL & CAUTION:**

GIS uses SF6 gas, reducing footprint by 85-90% compared to AIS. Slide 59 specifically showcases the KKU Indoor Substation as a prime example.

CH08-14 | Chapter 8 | Level 2 (Exam / Calc) | 📌 Ch08 Slide 7, 38 & Final Exam ข้อ 5.3

Transmission Towers, Tower Footing Resistance & Lightning Protection

📌 PROBLEM / QUESTION:

Classify high-voltage transmission towers into Suspension, Tension, Dead-end, and Transposition towers. Define Tower Footing Resistance (R_{tower}), state why it must be maintained $\leq 10\ \Omega$, and explain the Back Flashover phenomenon.



รูปประกอบ: High-Voltage Transmission Towers (Suspension, Tension, Transposition)

📐 KEY FORMULAS & DEFINITIONS:

Suspension (เสาแขวน): Straight route | Tension (เสารั้ง): Route angles/corners

Dead-end (เสาปลายสาย): Substation terminations | Transposition (เสาสลับเฟส): Phase balance

$$R_{\text{tower}} \leq 10\ \Omega, \qquad V_{\text{tower top}} = I_{\text{lightning}} \times R_{\text{tower}} + L\frac{di}{dt}$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Transmission Tower Classifications:
- Suspension Towers (เสาแขวน): Used along straight paths. Insulators hang vertically to support the physical weight of conductors.
- Tension / Angle Towers (เสารั้ง/เสามุม): Used at line bends and angles. Insulator strings are positioned horizontally under heavy mechanical tension.
- Dead-end Towers (เสาปลายสาย): Heavy-duty termination towers situated just outside substations to withstand full one-sided mechanical tension.
- Transposition Towers (เสาสลับเฟส): Rotates phase conductor spatial positions by 120 degrees at regular 1/3 distance intervals to balance line inductances, capacitances, and phase voltages.

2. Tower Footing Resistance (R_{tower}):
- Definition: The electrical contact resistance between the steel tower footing ground grid and the earth mass.
- Standard Requirement: Must be maintained strictly below $10\ \Omega$.

3. Back Flashover Phenomenon:
- When lightning strikes the overhead shield wire (OHGW) or tower peak, huge surge current ($I_{\text{lightning}} \approx 50 - 100\ \text{kA}$) flows down the steel tower to earth.
- Tower top voltage surges to $V_{\text{tower}} = I_{\text{lightning}} \times R_{\text{tower}}$.
- If $R_{\text{tower}} > 10\ \Omega$, V_{tower} exceeds the impulse flashover rating of the insulator strings, causing an electrical arc to jump from the tower back into the phase conductors (Back Flashover), tripping circuit breakers.

⚠️ EXAM PITFALL & CAUTION:

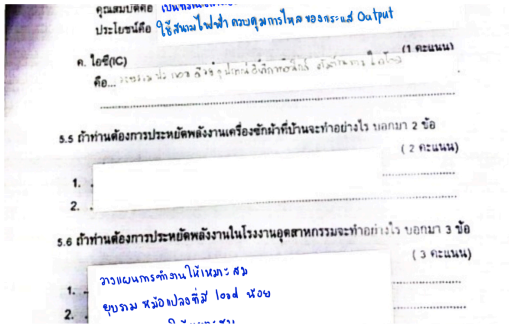
Back Flashover occurs from tower to line when footing resistance is too high ($> 10\ \Omega$). Keep $R_{\text{tower}} \leq 10\ \Omega$ with ground rods or counterpoise wires.

CH08-15 | Chapter 8 | Level 2 (Exam / Calc) | Final Exam ข้อ 5.5 & 5.6 (หน้า 24 - 8 คะแนนเต็ม)

Engineering Energy Conservation: Residential vs Industrial Systems

PROBLEM / QUESTION:

Propose 4 practical energy conservation measures for residential homes and 4 advanced engineering measures for industrial factories, detailing the underlying engineering principles (Power Factor correction, Inverter AC, VFD Affinity Laws).



รูปประกอบ: Final Exam Problems 5.5 & 5.6: Energy Conservation in Home and Factory

KEY FORMULAS & DEFINITIONS:

$$\text{Residential AC: } P = \frac{\text{BTU/hr}}{\text{SEER} \times 3.412} \implies \text{High SEER Inverter} + 26^\circ\text{C with Fan (Saves 30-40\%)}$$

$$\text{Industrial PF: } Q_C = P(\tan\theta_1 - \tan\theta_2) \implies \text{Eliminates Surcharges \& Reduces } I^2R$$

$$\text{Industrial Motor VFD: } P \propto N^3 \implies 20\% \text{ speed reduction cuts power by } 49\%!$$

STEP-BY-STEP MODEL SOLUTION:

1. Four Residential Energy Conservation Measures (Final Exam 5.5):
- 1) Inverter Air Conditioning Optimization: Select high SEER inverter units; operate at 25-26°C supplemented by electric fans. Cuts AC electricity consumption by 30-40%.
- 2) LED Lighting Conversion: Replace incandescent and fluorescent bulbs with LEDs, reducing lighting energy consumption by over 80% while reducing room heat load.
- 3) Thermal Insulation: Install ceiling glass-wool insulation and solar reflective roof paint to prevent heat transfer into living spaces.
- 4) Standby Power Elimination: Use master switch power strips to disconnect phantom loads (televisions, microwave ovens, chargers).

2. Four Industrial Energy Conservation Measures (Final Exam 5.6):
- 1) Power Factor (PF) Correction with Automatic Capacitor Banks: Elevates factory PF from ~0.75 to 0.95+. Eliminates reactive power penalty surcharges from PEA/MEA, reduces feeder I^2R loss, and frees transformer kVA capacity.
- 2) Variable Frequency Drives (VFD) on Pumps and Fans: Governed by Affinity Laws ($P \propto N^3$). Reducing pump or fan motor speed by 20% reduces required electrical power by $(0.8)^3 = 0.512$ (48.8% energy reduction!).
- 3) Waste Heat Recovery: Install heat exchangers and economizers on boiler flue gas and furnace exhausts to preheat boiler feed water.
- 4) Compressed Air System Optimization: Repair compressed air pipe leaks (typically wasting 20-30% of compressor energy), lower operating pressure, and install smart sequencing controllers.

EXAM PITFALL & CAUTION:

VFD Affinity Law: Power varies with the CUBE of speed ($P \propto N^3$). A 20% speed drop halves power consumption! PF correction saves penalties and I^2R losses.

PART 4

Chapter 9: Electrical Measuring Instruments & Sensors

PMMC Principles, DC Ammeters & Voltmeters with CDRX, Ayrton Shunts, Ohmmeters, Wattmeters, Accuracy (14 Flashcards)

CH09-01 | Chapter 9 | Level 1 (Core Concept) | 📌 Homework Chapter 10 & Final Exam ข้อ 3.1

PMMC / Galvanometer Working Principle & Deflection Torque

📌 **PROBLEM / QUESTION:**

Explain the operating principle of a Permanent Magnet Moving Coil (PMMC / D'Arsonval) instrument. State the formulas for deflecting torque and controlling torque, and explain why PMMC instruments measure only DC directly.

📐 **KEY FORMULAS & DEFINITIONS:**

$$T_d = B \cdot I \cdot N \cdot A = G \cdot I$$

$$T_c = C \cdot \theta$$

$$\text{At equilibrium: } T_d = T_c \implies \theta = \left(\frac{B \cdot N \cdot A}{C}\right) I \propto I$$

$$I_{\text{avg, AC}} = \frac{1}{T}\int_0^T I_m \sin(\omega t)\, dt = 0$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Operating Principle:

A rectangular coil with N turns and cross-sectional area A carrying current I is suspended in the uniform radial magnetic field B of a permanent magnet. The interaction between current and magnetic flux produces a Lorentz force ($F = ILB$), resulting in a deflecting torque T_d:

$$T_d = B \cdot I \cdot N \cdot A = G \cdot I$$

where $G = BNA$ is the galvanometer constant.

2. Controlling Torque & Linear Scale:

A spiral hairspring produces an opposing restoring torque T_c:

$$T_c = C \cdot \theta$$

where C is the spring constant and θ is the angular deflection. At equilibrium ($T_d = T_c$):

$$\theta = \left(\frac{B \cdot N \cdot A}{C}\right) I = \left(\frac{G}{C}\right) I$$

Because B, N, A, and C are constants, the pointer deflection angle θ is directly proportional to current I ($\theta \propto I$). This guarantees an evenly spaced, completely linear scale.

3. Why Measures Only DC Directly:

If connected to sinusoidal alternating current ($i(t) = I_m \sin \omega t$), torque alternates direction every half-cycle (50 Hz). Due to the mechanical inertia of the moving coil assembly, the pointer cannot track 50 Hz oscillations and responds strictly to the time-average value:

$$I_{\text{avg}} = \frac{1}{T}\int_0^T I_m \sin(\omega t)\, dt = 0$$

The pointer vibrates negligibly around zero. To measure AC, a diode rectifier must be placed before the PMMC movement.

⚠️ **EXAM PITFALL & CAUTION:**

PMMC measures strictly average DC. To measure AC with a PMMC meter, a rectifier circuit (diodes) must be added beforehand.

CH09-02 | Chapter 9 | Level 2 (Exam / Calc) | Homework Chapter 10 ข้อ 1 (Official Problem 1)

PMMC Torque & Deflection Calculation

PROBLEM / QUESTION:

A moving coil instrument has N = 150 turns, coil area A = 100 mm^2, and operates in a magnetic flux density B = 0.15 T. The spring constant is C = 2 × 10^-6 N*m/degree. Calculate the deflecting torque Td and pointer deflection angle theta when carrying a current of I = 5 mA.

KEY FORMULAS & DEFINITIONS:

$$A = 100\ \text{mm}^2 = 100 \times 10^{-6}\ \text{m}^2 = 10^{-4}\ \text{m}^2$$

$$T_d = B \cdot I \cdot N \cdot A$$

$$\theta = \frac{T_d}{C}$$

STEP-BY-STEP MODEL SOLUTION:

1. Convert Coil Area to SI Units (m^2):

$$A = 100\ \text{mm}^2 \times \left(\frac{1\ \text{m}}{1000\ \text{mm}}\right)^2 = 100 \times 10^{-6}\ \text{m}^2 = 10^{-4}\ \text{m}^2$$

2. Calculate Deflecting Torque (T_d):

$$T_d = B \cdot I \cdot N \cdot A$$

$$T_d = (0.15\ \text{T}) \times (5 \times 10^{-3}\ \text{A}) \times 150 \times (10^{-4}\ \text{m}^2)$$

$$T_d = 1.125 \times 10^{-5}\ \text{N} \cdot \text{m}$$

3. Calculate Pointer Deflection Angle (θ):

At steady deflection equilibrium ($T_d = T_c = C \cdot \theta$):

$$\theta = \frac{T_d}{C} = \frac{1.125 \times 10^{-5}\ \text{N} \cdot \text{m}}{2 \times 10^{-6}\ \text{N} \cdot \text{m/deg}} = 5.625^\circ$$

EXAM PITFALL & CAUTION:

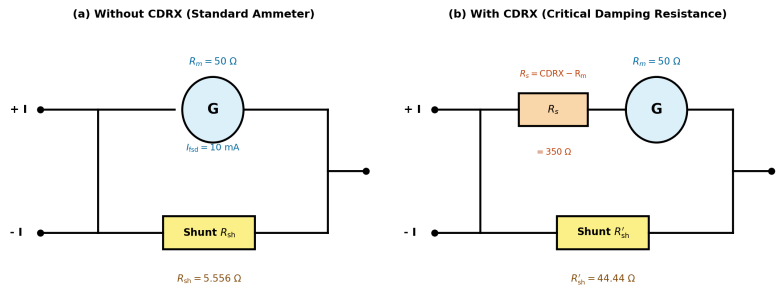
Always convert dimensions from mm^2 or cm^2 to m^2 (100 mm^2 = 10^-4 m^2). Failure to convert causes error by 1,000,000x.

CH09-03 | Chapter 9 | Level 2 (Exam / Calc) | Homework Chapter 10 ข้อ 2 (Official Problem 2)

DC Ammeter Design with and without CDRX

PROBLEM / QUESTION:

A Galvanometer has internal resistance Rg = 50 ohms, full-scale current Ig = 10 mA, and Critical Damping Resistance CDRX = 400 ohms. Design a DC Ammeter to measure full scale I = 100 mA for:
(a) Pure circuit without CDRX
(b) Circuit with CDRX (damping resistor).



รูปประกอบ: Homework 10 Problem 2: DC Ammeter Design with and without CDRX

KEY FORMULAS & DEFINITIONS:

$$\text{Without CDRX: } R_{\text{sh}} = \frac{I_g R_g}{I - I_g} = \frac{R_g}{m-1}$$

$$\text{With CDRX: Series resistor } R_s = \text{CDRX} - R_g = 400 - 50 = 350\ \Omega$$

$$R_{\text{branch}} = R_g + R_s = \text{CDRX} = 400\ \Omega$$

$$R'_{\text{sh}} = \frac{I_g \cdot R_{\text{branch}}}{I - I_g} = \frac{I_g \cdot \text{CDRX}}{I - I_g}$$

$$R_{\text{in}} = R'_{\text{sh}} \parallel R_{\text{branch}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Without CDRX:
Shunt current bypassing the coil:

$$I_{\text{sh}} = I - I_g = 100\ \text{mA} - 10\ \text{mA} = 90\ \text{mA} = 0.090\ \text{A}$$

Shunt resistor R_{sh}:

$$R_{\text{sh}} = \frac{I_g \cdot R_g}{I - I_g} = \frac{(0.010\ \text{A})(50\ \Omega)}{0.090\ \text{A}} = \frac{0.50}{0.090} = 5.556\ \Omega$$

2. Part (b) With Critical Damping Resistance (CDRX = 400 ohms):
To ensure critical damping in an ammeter, insert a series damping resistor R_s in the galvanometer branch:

$$R_s = \text{CDRX} - R_g = 400\ \Omega - 50\ \Omega = 350\ \Omega$$

Effective meter branch resistance:

$$R_{\text{branch}} = R_g + R_s = 400\ \Omega$$

New shunt resistor R'_{sh}:

$$R'_{\text{sh}} = \frac{I_g \cdot R_{\text{branch}}}{I - I_g} = \frac{(0.010\ \text{A})(400\ \Omega)}{0.090\ \text{A}} = \frac{4.0}{0.090} = 44.444\ \Omega$$

Total ammeter input resistance:

$$R_{\text{in}} = R'_{\text{sh}} \parallel R_{\text{branch}} = \frac{(44.444)(400)}{44.444 + 400} = 40.0\ \Omega$$

⚠️ **EXAM PITFALL & CAUTION:**

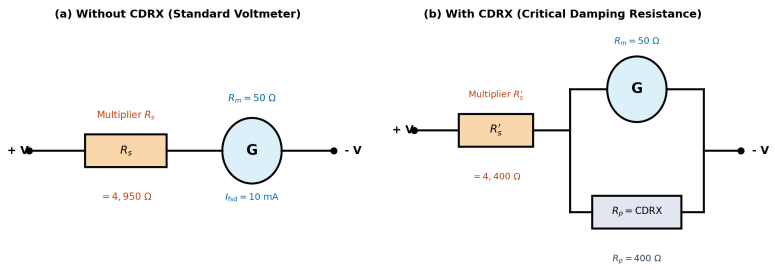
In Ammeters, CDRX requires a SERIES resistor Rs. Do not connect CDRX in parallel with Rg!

CH09-04 | Chapter 9 | Level 2 (Exam / Calc) | Homework Chapter 10 ข้อ 3 (Official Problem 3)

DC Voltmeter Design with and without CDRX

PROBLEM / QUESTION:

A Galvanometer has internal resistance Rg = 50 ohms, full-scale current Ig = 10 mA, and Critical Damping Resistance CDRX = 400 ohms. Design a DC Voltmeter to measure full scale V = 50 V for:
(a) Pure circuit without CDRX
(b) Circuit with CDRX (damping resistor).



รูปประกอบ: Homework 10 Problem 3: DC Voltmeter Design with and without CDRX

KEY FORMULAS & DEFINITIONS:

$$\text{Without CDRX: } R_s = \frac{V}{I_g} - R_g$$

$$\text{With CDRX: Parallel damping resistor } R_p = \text{CDRX} = 400\ \Omega$$

$$V_m = I_g \cdot R_g \qquad I_p = \frac{V_m}{R_p}$$

$$I_{\text{total}} = I_g + I_p = I_g \left(1 + \frac{R_g}{R_p}\right)$$

$$R'_s = \frac{V - V_m}{I_{\text{total}}}$$

$$\text{Sensitivity: } S = \frac{1}{I_{\text{total}}}\ (\Omega/\text{V})$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Without CDRX:
Total circuit resistance required:

$$R_{\text{total}} = \frac{V}{I_g} = \frac{50\ \text{V}}{0.010\ \text{A}} = 5,000\ \Omega = 5\ \text{k}\Omega$$

Multiplier series resistor R_s:

$$R_s = R_{\text{total}} - R_g = 5,000\ \Omega - 50\ \Omega = 4,950\ \Omega = 4.95\ \text{k}\Omega$$

2. Part (b) With Critical Damping Resistance (CDRX = 400 ohms):
For a voltmeter, connect damping resistor $R_p = \text{CDRX} = 400\ \Omega$ in PARALLEL across the meter movement:
- Voltage drop across coil at full scale:

$$V_m = I_g \cdot R_g = (0.010\ \text{A})(50\ \Omega) = 0.5\ \text{V}$$

- Current drawn by parallel damping resistor R_p:

$$I_p = \frac{V_m}{R_p} = \frac{0.5\ \text{V}}{400\ \Omega} = 1.25\ \text{mA} = 0.00125\ \text{A}$$

- Total current entering the voltmeter:

$$I_{\text{total}} = I_g + I_p = 10\ \text{mA} + 1.25\ \text{mA} = 11.25\ \text{mA} = 0.01125\ \text{A}$$

- Multiplier series resistor R_s':

$$R_s' = \frac{V - V_m}{I_{\text{total}}} = \frac{50\ \text{V} - 0.5\ \text{V}}{0.01125\ \text{A}} = \frac{49.5\ \text{V}}{0.01125\ \text{A}} = 4,400\ \Omega = 4.4\ \text{k}\Omega$$

- Voltmeter sensitivity:

$$S = \frac{1}{I_{\text{total}}} = \frac{1}{0.01125\ \text{A}} = 88.89\ \Omega/\text{V}$$

⚠ **EXAM PITFALL & CAUTION:**

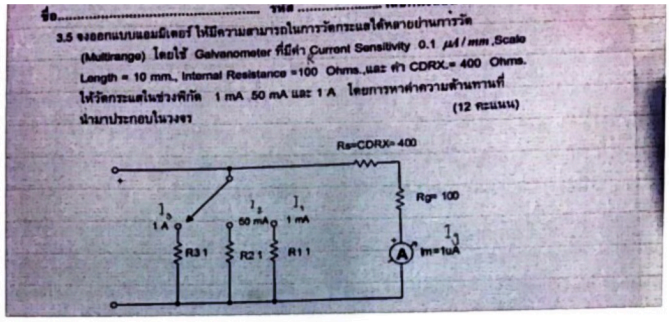
In Ammeters, CDRX requires a SERIES resistor Rs. In Voltmeters, CDRX requires a PARALLEL resistor Rp! Remember this contrast.

CH09-05 | Chapter 9 | Level 2 (Exam / Calc) | Final Exam ข้อ 3.5 (หน้า 29 - 6 คะแนน)

Multi-Range DC Ammeter Design with CDRX

PROBLEM / QUESTION:

A galvanometer has sensitivity 0.1 uA/mm, full scale deflection 10 mm, coil resistance Rg = 100 ohms, and critical damping resistance CDRX = 400 ohms. Design a 3-range DC ammeter for I1 = 1 mA, I2 = 50 mA, and I3 = 1 A using individual shunts. Determine damping resistor Rs and shunt resistors R11, R21, R31.



รูปประกอบ: Exam Problem 3.5: Multi-Range DC Ammeter with CDRX

KEY FORMULAS & DEFINITIONS:

$$I_{\text{fsd}} = \text{Sensitivity} \times \text{Scale Length} = (0.1\ \mu\text{A/mm})(10\ \text{mm}) = 1\ \mu\text{A} = 10^{-6}\ \text{A}$$

$$R_s = \text{CDRX} - R_g = 400\ \Omega - 100\ \Omega = 300\ \Omega$$

$$R_{\text{branch}} = R_g + R_s = 400\ \Omega$$

$$R_{n1} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_n - I_{\text{fsd}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Determine Full-Scale Galvanometer Current (I_{fsd}):

$$I_{\text{fsd}} = 0.1\ \mu\text{A/mm} \times 10\ \text{mm} = 1\ \mu\text{A} = 10^{-6}\ \text{A}$$

2. Determine Series Damping Resistor (R_s):

$$R_s = \text{CDRX} - R_g = 400\ \Omega - 100\ \Omega = 300\ \Omega$$

Total meter branch resistance:

$$R_{\text{branch}} = R_g + R_s = 400\ \Omega$$

3. Calculate Shunt Resistors for Each Range:

- Range 1 ($I_1 = 1\ \text{mA} = 10^{-3}\ \text{A}$):

$$R_{11} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_1 - I_{\text{fsd}}} = \frac{(10^{-6}\ \text{A})(400\ \Omega)}{10^{-3}\ \text{A} - 10^{-6}\ \text{A}} = \frac{0.0004}{0.000999} = 0.4004\ \Omega$$

- Range 2 ($I_2 = 50\ \text{mA} = 0.05\ \text{A}$):

$$R_{21} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_2 - I_{\text{fsd}}} = \frac{(10^{-6}\ \text{A})(400\ \Omega)}{0.05\ \text{A} - 10^{-6}\ \text{A}} = \frac{0.0004}{0.049999} = 0.00800\ \Omega$$

- Range 3 ($I_3 = 1\ \text{A}$):

$$R_{31} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_3 - I_{\text{fsd}}} = \frac{(10^{-6}\ \text{A})(400\ \Omega)}{1.0\ \text{A} - 10^{-6}\ \text{A}} = \frac{0.0004}{0.999999} = 0.000400\ \Omega$$

EXAM PITFALL & CAUTION:

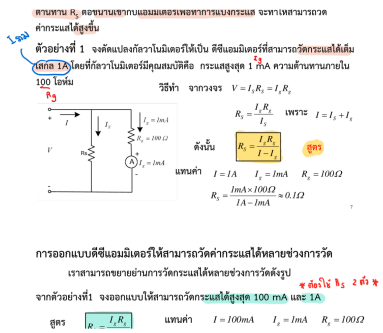
Notice that higher current range requires SMALLER shunt resistance (R31 < R21 < R11) because more current must be bypassed around the meter movement.

CH09-06 | Chapter 9 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Ayrton Shunt (Universal Shunt) for Multi-Range Ammeter

PROBLEM / QUESTION:

What is an Ayrton Shunt (Universal Shunt)? Why is it superior to individual switched shunts, and how do you calculate the resistor values Ra, Rb, and Rc for multi-range operation?



รูปประกอบ: Ayrton / Universal Shunt Multi-Range Ammeter Circuit

KEY FORMULAS & DEFINITIONS:

$$\text{Total Shunt: } R_{\text{sh}} = R_a + R_b + R_c$$

$$\text{Range 1 (Lowest } I_1\text{): } R_{\text{sh}} = \frac{I_{\text{fsd}} R_m}{I_1 - I_{\text{fsd}}}$$

$$\text{Range 2 (Medium } I_2\text{): } (R_b + R_c) = \frac{I_{\text{fsd}}(R_m + R_a)}{I_2}$$

$$\text{Range 3 (Highest } I_3\text{): } R_c = \frac{I_{\text{fsd}}(R_m + R_a + R_b)}{I_3}$$

STEP-BY-STEP MODEL SOLUTION:

1. What is an Ayrton Shunt:
A tapped series chain of resistors ($R_a + R_b + R_c$) permanently connected in parallel across the meter movement R_m. Switching ranges moves the circuit input tap, altering which resistors act as shunt and which are added in series with the meter coil.

2. Decisive Advantage Over Individual Shunts:
- Make-Before-Break Safety: In individual shunts, if the rotary switch momentarily opens between contacts, the entire line current would blast through the fragile meter movement (R_m), destroying it instantly.
- In Ayrton shunt, the meter movement is NEVER left open without a parallel bypass path during range switching.

3. Equations for 3 Ranges:
- Lowest current I_1:

$$R_a + R_b + R_c = \frac{I_{\text{fsd}} R_m}{I_1 - I_{\text{fsd}}}$$

- Medium current I_2:

$$R_b + R_c = \frac{I_{\text{fsd}}(R_m + R_a)}{I_2}$$

- Highest current I_3:

$$R_c = \frac{I_{\text{fsd}}(R_m + R_a + R_b)}{I_3}$$

EXAM PITFALL & CAUTION:

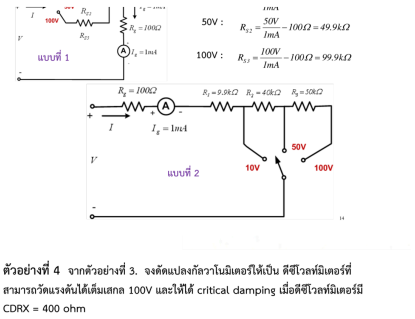
In Ayrton shunt calculations, the resistors not acting as shunt become added in series with Rm! Keep track of which resistors are in parallel vs in series.

CH09-07 | Chapter 9 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Multi-Range Voltmeter (Individual vs Tapped Series)

PROBLEM / QUESTION:

A meter has Rm = 100 ohms and Ifsd = 50 microamperes. Calculate the multiplier resistor values R1, R2, and R3 to construct a multi-range voltmeter for 10 V, 50 V, and 250 V using a tapped series multiplier configuration.



รูปประกอบ: Tapped Series Multiplier Multi-Range Voltmeter

KEY FORMULAS & DEFINITIONS:

$$S = \frac{1}{I_{\text{fsd}}} = \frac{1}{50 \times 10^{-6}\ \text{A}} = 20,000\ \Omega/\text{V} \quad (20\ \text{k}\Omega/\text{V})$$

$$R_{\text{total},1} = S \cdot V_1 = (20\ \text{k}\Omega/\text{V})(10\ \text{V}) = 200\ \text{k}\Omega$$

$$R_1 = R_{\text{total},1} - R_m = 200\ \text{k}\Omega - 100\ \Omega = 199.9\ \text{k}\Omega$$

$$R_2 = S(V_2 - V_1) = 20\ \text{k}\Omega/\text{V} \cdot (50 - 10) = 800\ \text{k}\Omega$$

$$R_3 = S(V_3 - V_2) = 20\ \text{k}\Omega/\text{V} \cdot (250 - 50) = 4,000\ \text{k}\Omega = 4\ \text{M}\Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Voltmeter Sensitivity (S):

$$S = \frac{1}{I_{\text{fsd}}} = \frac{1}{50 \times 10^{-6}\ \text{A}} = 20,000\ \Omega/\text{V} = 20\ \text{k}\Omega/\text{V}$$

2. Range 1 ($V_1 = 10\ \text{V}$):

$$R_{\text{total},1} = S \cdot V_1 = (20\ \text{k}\Omega/\text{V})(10\ \text{V}) = 200\ \text{k}\Omega$$

$$R_1 = R_{\text{total},1} - R_m = 200\ \text{k}\Omega - 0.1\ \text{k}\Omega = 199.9\ \text{k}\Omega$$

3. Range 2 ($V_2 = 50\ \text{V}$):
In a tapped series chain, each incremental resistor spans the voltage difference between adjacent ranges:

$$R_2 = S \cdot (V_2 - V_1) = (20\ \text{k}\Omega/\text{V})(50\ \text{V} - 10\ \text{V}) = (20\ \text{k}\Omega/\text{V})(40\ \text{V}) = 800\ \text{k}\Omega$$

4. Range 3 ($V_3 = 250\ \text{V}$):

$$R_3 = S \cdot (V_3 - V_2) = (20\ \text{k}\Omega/\text{V})(250\ \text{V} - 50\ \text{V}) = (20\ \text{k}\Omega/\text{V})(200\ \text{V}) = 4,000\ \text{k}\Omega = 4\ \text{M}\Omega$$

EXAM PITFALL & CAUTION:

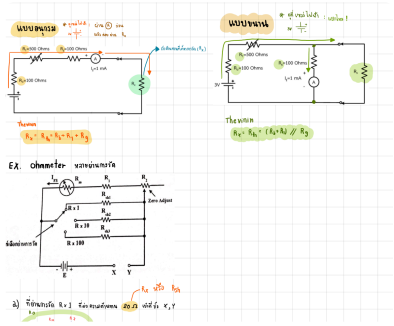
In a tapped series chain, each higher resistor R_n is simply S * (V_n - V_{n-1}), not the full resistance.

CH09-08 | Chapter 9 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam ข้อ 3

Series-Type Ohmmeter Working Principle & Scale

PROBLEM / QUESTION:

Explain the circuit design of a Series-Type Ohmmeter. Describe its pointer deflection at Rx = 0, Rx = R_half, and Rx = infinity, and explain why its resistance scale is non-linear and reads backwards.



รูปประกอบ: Series-Type Ohmmeter Circuit & Scale

KEY FORMULAS & DEFINITIONS:

$$I = \frac{E}{R_{\text{in}} + R_x}$$

$$R_x = 0 \implies I = I_{\text{fsd}} = \frac{E}{R_{\text{in}}} \quad (\text{Far Right: } 0\ \Omega)$$

$$R_x = R_h = R_{\text{in}} \implies I = 0.5\, I_{\text{fsd}} \quad (\text{Center of scale: } R_h)$$

$$R_x = \infty \implies I = 0 \quad (\text{Far Left zero current: } \infty\ \Omega)$$

STEP-BY-STEP MODEL SOLUTION:

1. Circuit Topology:
An internal battery E, current-limiting resistor R_1, zero-adjust rheostat R_2, and galvanometer R_m are all connected in series with the unknown resistance R_x across test terminals.

2. Scale Deflection Points:
- $R_x = 0$ (Short Circuit): Maximum current flows, deflecting pointer to full-scale at the far RIGHT:

$$I = I_{\text{fsd}} = \frac{E}{R_{\text{in}}} \implies 0\ \Omega$$

- $R_x = R_{\text{in}} = R_h$ (Half-Scale Resistance): Total resistance doubles, halving current to exactly mid-scale:

$$I = \frac{E}{2R_{\text{in}}} = 0.5\, I_{\text{fsd}} \implies R_h$$

- $R_x = \infty$ (Open Circuit): No current flows, pointer remains at rest at far LEFT:

$$I = 0 \implies \infty\ \Omega$$

3. Why Scale is Non-Linear & Reads Backwards:
Because current $I = E/(R_{\text{in}} + R_x)$ varies inversely with R_x, deflection is hyperbolic. Markings are widely spaced on the right (low ohms) and heavily compressed on the left (high ohms).

EXAM PITFALL & CAUTION:

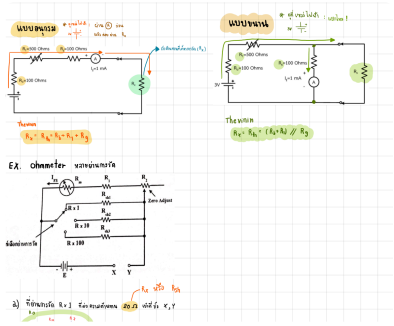
Series ohmmeter scale reads 0 on the RIGHT and infinity on the LEFT (opposite to voltmeters and ammeters).

CH09-09 | Chapter 9 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Shunt-Type Ohmmeter for Low Resistance Measurement

PROBLEM / QUESTION:

Explain the operating principle of a Shunt-Type Ohmmeter. For what range of resistance is it suited, and how does its scale direction differ from a series ohmmeter?



รูปประกอบ: Shunt-Type Ohmmeter Circuit & Scale

KEY FORMULAS & DEFINITIONS:

$$I_m = I_{\text{total}} \cdot \frac{R_x}{R_m + R_x}$$

$$R_x = 0 \implies I_m = 0 \quad (\text{Far Left edge} = 0\ \Omega)$$

$$R_x = R_m \implies I_m = 0.5\, I_{\text{total}} \quad (\text{Center of scale} = R_m)$$

$$R_x = \infty \implies I_m = I_{\text{fsd}} \quad (\text{Far Right edge} = \infty\ \Omega)$$

$$\text{Measurement range: Low resistance } (5\ \text{m}\Omega - 100\ \Omega)$$

STEP-BY-STEP MODEL SOLUTION:

1. Operating Principle:
The unknown resistor R_x is connected in PARALLEL (shunt) across the meter movement R_m, rather than in series. A series battery and resistor supply constant circuit current.

2. Scale Deflection Points:
- $R_x = 0$ (Short Circuit): All current bypasses through the zero-ohm short; coil current $I_m = 0$. Pointer stays at far LEFT = $0\ \Omega$:

$$I_m = 0 \implies 0\ \Omega$$

- $R_x = R_m$: Current splits equally; pointer deflects to mid-scale ($0.5 I_{\text{fsd}}$) = R_m.
- $R_x = \infty$ (Open Circuit): No current bypassed; full current flows through R_m, deflecting pointer to far RIGHT = $\infty\ \Omega$:

$$I_m = I_{\text{fsd}} \implies \infty\ \Omega$$

Result: Shunt ohmmeter scale reads from LEFT TO RIGHT (0 on left, ∞ on right), matching normal voltmeter/ammeter scale direction!

3. Primary Application:
Measuring very LOW resistances ($0.01\ \Omega$ to a few tens of ohms), such as motor winding coils, contact resistance, and transformer wire.

EXAM PITFALL & CAUTION:

Series ohmmeter: 0 is on the RIGHT. Shunt ohmmeter: 0 is on the LEFT. Always know which type the question asks!

CH09-10 | Chapter 9 | Level 1 (Core Concept) | 📌 Lecture Chapter 10 & Lab Multimeter

Multi-Range Ohmmeter & Zero-Adjust Rheostat Function

📌 **PROBLEM / QUESTION:**

What is the primary function of the Zero-Adjust Rheostat (Zero-Ohm Adjust) in an analog multimeter, and how do multi-range multipliers (Rx1, Rx10, Rx10k) scale the center-scale resistance Rh?

📐 **KEY FORMULAS & DEFINITIONS:**

$$R_{\text{measured}} = \text{Dial Reading} \times \text{Range Multiplier}$$

$$R_{h,\text{new}} = R_h \times \text{Multiplier}$$

$$\text{Battery Decline: } E \downarrow \implies \text{Decrease } R_{\text{zero}} \text{ so that } I_{\text{fsd}} = \frac{E}{R_{\text{in}}}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Purpose of Zero-Adjust Rheostat:
As the internal dry cell battery ages, its open-circuit terminal voltage E drops and its internal resistance rises. The Zero-Adjust potentiometer compensates for this battery decline by decreasing circuit resistance so that shorting test leads ($R_x = 0$) still yields exactly full-scale current I_{fsd} (pointer aligns perfectly to $0\ \Omega$).

2. Multi-Range Operation:
Switching to higher ranges (e.g., $R \times 1, R \times 10, R \times 100, R \times 1\text{k}, R \times 10\text{k}$) modifies internal shunt/series divider resistors, multiplying the center-scale half-deflection resistance R_h by that factor:

$$R_{h,\text{new}} = R_h \times \text{Multiplier}$$

To measure $100\ \text{k}\Omega$ accurately, select $R \times 10\text{k}$ so the pointer deflects near center scale (R_h) where reading precision is highest.

⚠️ **EXAM PITFALL & CAUTION:**

Always re-zero the meter (touch leads together and adjust 0-ohm knob) EVERY TIME you switch resistance ranges!

CH09-11 | Chapter 9 | Level 2 (Exam / Calc) | Final Exam ข้อ 3.2 (หน้า 30 - 3 คะแนน)

Rectifier AC Voltmeter & Form Factor

PROBLEM / QUESTION:

A PMMC instrument is used with a full-wave bridge rectifier to measure sinusoidal AC voltage. Derive the relationship between the meter's DC response and the AC RMS voltage, and explain what happens if a square wave or triangular wave is measured.

KEY FORMULAS & DEFINITIONS:

$$V_{\text{avg}} = \frac{2}{\pi} V_m \approx 0.637\, V_m$$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \approx 0.707\, V_m$$

$$\text{Form Factor } k_f = \frac{V_{\text{rms}}}{V_{\text{avg}}} = \frac{0.707\, V_m}{0.637\, V_m} = 1.11$$

$$V_{\text{scale}} = 1.11 \times V_{\text{avg}} = V_{\text{rms}}$$

STEP-BY-STEP MODEL SOLUTION:

1. PMMC Mechanism Response:
The moving coil responds strictly to average rectified DC current:

$$V_{\text{avg}} = \frac{2}{\pi} V_m \approx 0.637\, V_m$$

2. AC Calibration (Form Factor):
Users require the dial to display Effective / Root-Mean-Square (RMS) voltage. For pure sinusoidal AC, the Form Factor is:

$$k_f = \frac{V_{\text{rms}}}{V_{\text{avg}}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}} \approx 1.11$$

Manufacturers calibrate dial markings by multiplying average deflection by 1.11:

$$V_{\text{dial}} = 1.11 \times V_{\text{avg}} = V_{\text{rms}}$$

3. Non-Sinusoidal Waveform Error:
For a square wave, $k_f = 1.00$. Measuring a 10 V peak square wave causes the meter to read $1.11 \times 10 = 11.1$ V (an error of $+11\%$)! For non-sinusoidal waveforms, True-RMS meters must be utilized.

EXAM PITFALL & CAUTION:

Standard analog AC voltmeters assume a pure sine wave (kf = 1.11). Measuring a 10 V peak square wave will read 11.1 V instead of 10.0 V RMS!

CH09-12 | Chapter 9 | Level 1 (Core Concept) | Final Exam ข้อ 3.4 (หน้า 30 - 3 คะแนน)

Wattmeter Connection to Load (Current vs Potential Coils)

PROBLEM / QUESTION:

Describe the construction of an Electrodynamometer Wattmeter. How are the Current Coil (CC) and Potential/Voltage Coil (PC) connected to a load, and what precautions prevent coil burnout?



รูปประกอบ: Exam Problem 3.4: Electrodynamometer Wattmeter Connection to Load

KEY FORMULAS & DEFINITIONS:

$$P = V_{\text{load}} \cdot I_{\text{load}} \cdot \cos\theta$$

$$\text{Current Coil (CC): Fixed, thick wire, low turns} \implies \text{Connected in SERIES with load (Terminals 1–2)}$$

$$\text{Potential Coil (PC): Moving, fine wire, high turns} \implies \text{Connected in PARALLEL across load (Terminals 3–5)}$$

$$\text{Burnout Hazard: Connecting CC in parallel creates a dead short circuit } I_{\text{sc}} = \frac{V}{R_{\text{CC}}} \approx \infty$$

STEP-BY-STEP MODEL SOLUTION:

1. Wattmeter Coils & Connection:
- Current Coil (CC / ขดลวดกระแส): Made of thick copper wire with few turns (near-zero impedance). MUST be connected in SERIES with the load (terminals 1-2) to carry full line current.
- Potential Coil (PC / ขดลวดแรงดัน): Made of fine wire with many turns, connected in series with a high non-inductive multiplier resistance. MUST be connected in PARALLEL across the load terminals (terminals 3-5) to measure voltage.

2. Power Indication:
Instantaneous torque is proportional to the product of magnetic fields produced by both coils: $T_d \propto i_{\text{cc}}(t) \cdot i_{\text{pc}}(t)$, deflecting the pointer to indicate true real average active power:

$$P = V_{\text{load}} \cdot I_{\text{load}} \cdot \cos\theta$$

3. Burnout Hazard:
NEVER connect the Current Coil (CC) in parallel across the voltage supply! Because CC has near-zero resistance, connecting it in parallel creates an immediate dead short circuit ($I_{\text{sc}} = V/R_{\text{CC}} \approx \infty$), vaporizing the coil instantly.

EXAM PITFALL & CAUTION:

Current Coil connects in SERIES. Potential Coil connects in PARALLEL. Reversing this causes explosive short circuits or an inoperative meter.

CH09-13 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 3.3 (หน้า 30 - 2 คะแนน)

Energy Meter (Kilowatt-Hour Meter) Function

📌 **PROBLEM / QUESTION:**

Explain the working principle of an induction-type Kilowatt-Hour Meter (kWh meter). What creates the driving torque on the aluminum disc, and what is the role of the permanent brake magnet?

📐 **KEY FORMULAS & DEFINITIONS:**

$$T_d \propto V \cdot I \cdot \cos\theta = P$$

$$T_b \propto \omega \quad \text{(Eddy-current braking)}$$

$$\text{At steady speed: } T_d = T_b \implies \omega \propto P$$

$$\text{Total Revolutions } N = \int \omega \, dt \propto \int P \, dt = \text{Energy (kWh)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Driving Torque Generation:

The meter contains a shunt electromagnet (connected across line voltage) and a series electromagnet (carrying load current). Their alternating magnetic fluxes induce eddy currents inside a lightweight aluminum disc. The interaction between each eddy current and the alternating flux produces a continuous rotational driving torque T_d proportional to instantaneous real power P:

$$T_d \propto V \cdot I \cdot \cos\theta = P$$

2. Braking Magnet Role:

A C-shaped permanent magnet sits near the outer edge of the disc. As the disc spins through this stationary field, eddy currents are induced in the disc, creating a retarding braking torque T_b directly proportional to rotational speed ω:

$$T_b \propto \omega$$

3. Energy Integration:

Equating driving and braking torque ($T_d = T_b$) yields:

$$\omega \propto P$$

The total number of disc revolutions N over time is directly proportional to cumulative electrical energy in kilowatt-hours (kWh):

$$N = \int \omega \, dt \propto \int P \, dt = \text{Energy (kWh)}$$

⚠️ **EXAM PITFALL & CAUTION:**

The disc speed is proportional to POWER (kW), but the total number of revolutions is proportional to ENERGY (kWh).

CH09-14 | Chapter 9 | Level 1 (Core Concept) | Lecture Chapter 10 & Lab Rules

Instrument Accuracy, Precision, and Error Formulas

PROBLEM / QUESTION:

Define the terms Accuracy, Precision, Absolute Error, Relative Error, and Percentage Accuracy in electrical measurement.

KEY FORMULAS & DEFINITIONS:

$$\text{Absolute Error: } e = Y_n - X_n$$

$$\text{Relative Error: } e_r = \frac{Y_n - X_n}{X_n}$$

$$\%\ \text{Error} = \frac{|Y_n - X_n|}{X_n} \times 100\%$$

$$\%\ \text{Accuracy } A = 100\% - \%\ \text{Error} = \left(1 - \frac{|Y_n - X_n|}{X_n}\right) \times 100\%$$

STEP-BY-STEP MODEL SOLUTION:

1. Accuracy (ความแม่นยำ/ความถูกต้อง):
How close the measured reading Y_n is to the true, accepted standard value X_n.

2. Precision (ความเที่ยงตรง):
The repeatability or consistency of successive measurements of the exact same physical quantity under identical conditions, regardless of whether it is near the true value.

3. Error & Accuracy Formulas:
- Absolute Error:

$$e = Y_n - X_n$$

- Relative Error:

$$e_r = \frac{Y_n - X_n}{X_n}$$

- Percentage Error:

$$\%\ \text{Error} = \frac{|Y_n - X_n|}{X_n} \times 100\%$$

- Percentage Accuracy (A):

$$A = 100\% - \%\ \text{Error} = \left(1 - \frac{|Y_n - X_n|}{X_n}\right) \times 100\%$$

EXAM PITFALL & CAUTION:

An instrument can have high precision (reads 9.81 V every single time) but low accuracy (if true voltage is 12.00 V).

PART 5

Chapter 10: Semiconductor & Electronic Devices

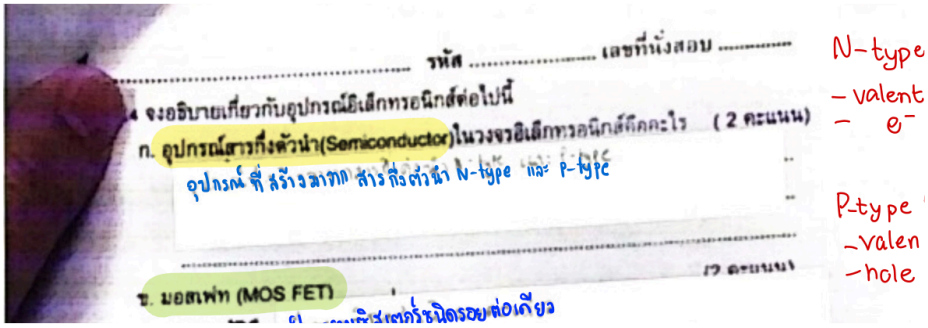
Intrinsic/Extrinsic, PN Diodes, Zener Regulators, Solar Cells, BJTs, MOSFETs, Rectifiers, Op-Amps (12 Flashcards)

CH10-01 | Chapter 10 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.1 (หน้า 24 - 2 คะแนน)

Intrinsic vs Extrinsic, N-Type vs P-Type Semiconductors

📌 **PROBLEM / QUESTION:**

Distinguish between Intrinsic and Extrinsic semiconductors. Explain how N-Type and P-Type semiconductors are formed by doping, identifying the majority and minority charge carriers in each.



รูปประกอบ: Exam Problem 4.1: Intrinsic & Extrinsic Semiconductors

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{N-Type: Si (4 val)} + \text{Pentavalent (5 val: P, As, Sb)} \implies \text{Majority: Free Electrons } (e^-)$$

$$\text{P-Type: Si (4 val)} + \text{Trivalent (3 val: B, Ga, In)} \implies \text{Majority: Holes } (h^+)$$

$$\text{Mass Action Law: } n \cdot p = n_i^2$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Intrinsic vs Extrinsic Semiconductors:
- Intrinsic: Pure semiconductor crystal (Silicon, valence = 4) with no impurities. Low conductivity at room temperature.
- Extrinsic: Doped with tiny concentrations of impurity atoms to dramatically increase conductivity.

2. N-Type Semiconductor:
- Doped with Pentavalent impurities (valence 5: Phosphorus, Arsenic, Antimony).
- 4 valence electrons bond with Silicon; the 5th electron becomes a free conduction electron.
- Majority Carriers: Free Electrons (e^-); Minority Carriers: Holes (h^+).

3. P-Type Semiconductor:
- Doped with Trivalent impurities (valence 3: Boron, Gallium, Indium).
- 3 valence electrons bond with Silicon, leaving a vacancy or 'hole' (h^+).
- Majority Carriers: Holes (h^+); Minority Carriers: Electrons (e^-).

4. Mass Action Law:

$$n \cdot p = n_i^2$$

⚠️ **EXAM PITFALL & CAUTION:**

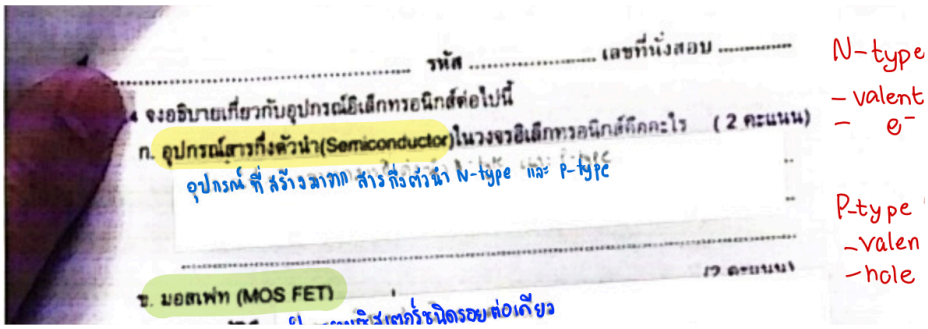
Both N-type and P-type semiconductors remain overall ELECTRICALLY NEUTRAL because total protons equal total electrons in the crystal lattice.

CH10-02 | Chapter 10 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.2 (หน้า 24 - 2 คะแนน)

MOSFET Definition, Properties & Uses

📌 **PROBLEM / QUESTION:**

What does MOSFET stand for? Describe its 3-terminal structure, explain why its gate input impedance is near-infinite, and state its primary advantages over BJT transistors.



รูปประกอบ: Exam Problem 4.2: MOSFET Definition & Characteristics

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{MOSFET} = \text{Metal-Oxide-Semiconductor Field-Effect Transistor}$$

$$\text{Terminals: Gate (G), Drain (D), Source (S)}$$

$$I_G \approx 0 \implies R_{\text{in, gate}} \approx 10^{12} - 10^{14}\ \Omega$$

$$I_D = K(V_{GS} - V_{th})^2 \quad \text{(Voltage-Controlled Device)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Definition & Terminals:
MOSFET stands for Metal-Oxide-Semiconductor Field-Effect Transistor. Its 3 active terminals are Gate (G), Drain (D), and Source (S).

2. Near-Infinite Gate Input Impedance:
The Gate electrode is physically isolated from the conducting channel by an ultra-thin dielectric layer of Silicon Dioxide (SiO_2). Because SiO_2 is an outstanding electrical insulator, static gate current is practically zero:

$$I_G \approx 0 \implies R_{\text{in, gate}} \approx 10^{12} - 10^{14}\ \Omega$$

3. Key Advantages Over BJT:
- Voltage-Controlled: Requires zero continuous gate current to stay ON, unlike BJT which demands base current I_B.
- High Switching Speed: Fast transition times ideal for high-frequency switch-mode power supplies and CPU logic gates.
- Positive Thermal Coefficient: Prevents thermal runaway and allows straightforward parallel connection.

⚠️ **EXAM PITFALL & CAUTION:**

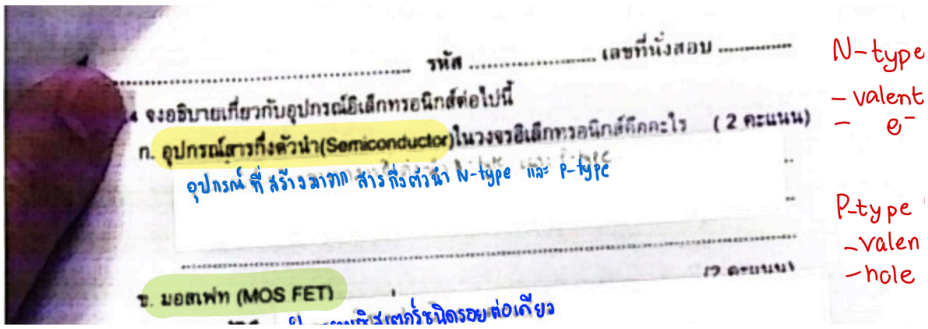
MOSFET is voltage-controlled (Vgs controls Id). BJT is current-controlled (Ib controls Ic).

CH10-03 | Chapter 10 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.3 (หน้า 24 - 2 คะแนน)

Integrated Circuit (IC) Definition & Significance

📌 **PROBLEM / QUESTION:**

What is an Integrated Circuit (IC)? How is it manufactured on a single semiconductor wafer, and what revolutionary benefits does it provide over discrete component circuits?



รูปประกอบ: Exam Problem 4.3: Integrated Circuit (IC) Significance

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{IC} = \text{Microscopic integration of Transistors, Diodes, Resistors on a single Silicon chip}$$

$$\text{Moore's Law: Transistor density doubles approximately every 2 years}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Definition of an Integrated Circuit (IC):
An Integrated Circuit (IC or microchip) is a complete electronic circuit consisting of millions to billions ($> 10^9$) of interconnected microscopic components (transistors, diodes, resistors, capacitors) fabricated monolithically on a single tiny semiconductor die (Silicon wafer):

$$\text{Microscopic Integration: Transistors } (> 10^9) + \text{Interconnects on single Si die}$$

2. Revolutionary Advantages over Discrete Circuits:
- Extreme Miniaturization: Shrinks entire computers onto a thumbnail-sized die ($< 1\ \text{cm}^2$).
- Low Unit Cost: Mass-produced simultaneously by the thousands on silicon wafers using optical photolithography.
- High Reliability: Monolithic metal layers replace thousands of fragile soldered wire joints.
- High Speed & Low Power: Nanometer trace distances allow gigahertz clock rates with negligible parasitic delays ($RC \to 0$).

⚠️ **EXAM PITFALL & CAUTION:**

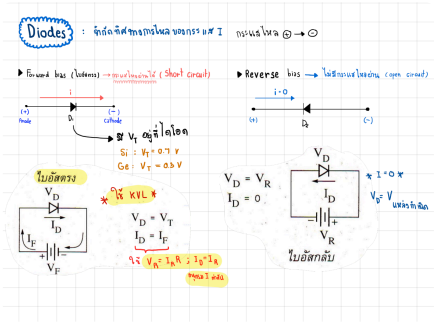
ICs are monolithic (all components created simultaneously on the same crystal substrate), not soldered discrete parts.

CH10-04 | Chapter 10 | Level 1 (Core Concept) | 📌 Lecture Chapter 10 & Final Exam

PN Junction Diode Biasing & Knee Voltages

📌 **PROBLEM / QUESTION:**

Explain Forward and Reverse Biasing of a PN junction diode. State the barrier potential (Knee Voltage) for Silicon and Germanium diodes, and explain the mechanism of reverse breakdown.



รูปประกอบ: PN Junction Diode Characteristic Curve & Biasing

📐 **KEY FORMULAS & DEFINITIONS:**

$$V_{\text{knee, Si}} \approx 0.7 \text{ V} \qquad V_{\text{knee, Ge}} \approx 0.3 \text{ V}$$

$$\text{Forward Bias: } V_{\text{anode}} > V_{\text{cathode}} + V_{\text{knee}} \implies I_D \text{ conducts heavily}$$

$$\text{Reverse Bias: } V_{\text{cathode}} > V_{\text{anode}} \implies I_D \approx I_s \text{ (Reverse Saturation Current } \approx \text{nA)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Forward Bias (ไบแอสตรง):
Positive supply terminal connected to Anode (P-side), Negative to Cathode (N-side). When applied voltage overcomes the internal barrier potential (0.7 V for Silicon, 0.3 V for Germanium), the depletion layer collapses and large forward current I_D conducts exponentially.

2. Reverse Bias (ไบแอสกลับ):
Positive supply to Cathode (N-side), Negative to Anode (P-side). External voltage widens the depletion barrier, preventing majority carrier transit. Only an ultra-tiny temperature-dependent reverse saturation current I_s (picoamps to nanoamps) leaks through.

3. Reverse Breakdown:
If reverse voltage exceeds Peak Inverse Voltage (PIV), avalanche multiplication causes sudden massive reverse current. Unless current is externally limited by a resistor, the diode burns out instantly.

⚠️ **EXAM PITFALL & CAUTION:**

Memorize: Silicon diode Knee Voltage = 0.7 V; Germanium diode Knee Voltage = 0.3 V. Standard problems assume Silicon 0.7 V.

CH10-05 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Diode Shockley Equation, Dynamic Resistance & LED Circuit

PROBLEM / QUESTION:

State Shockley's ideal diode equation. Define dynamic AC resistance rd, and calculate the current-limiting resistor R required to operate an LED (V_led = 2.0 V, I_led = 15 mA) from a 12 V DC source.

KEY FORMULAS & DEFINITIONS:

$$I_D = I_s\left(e^{\frac{V_D}{nV_T}} - 1\right) \qquad V_T = \frac{kT}{q} \approx 26\ \text{mV at } 300\ \text{K}$$

$$r_d = \frac{nV_T}{I_D} \approx \frac{26\ \text{mV}}{I_D}$$

$$R = \frac{V_{CC} - V_{\text{LED}}}{I_{\text{LED}}} = \frac{12\ \text{V} - 2.0\ \text{V}}{15 \times 10^{-3}\ \text{A}} = 667\ \Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Shockley Diode Equation:

$$I_D = I_s\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

where I_s is reverse saturation current, n is ideality factor (1 to 2), and $V_T = kT/q \approx 26\ \text{mV}$ at room temperature ($300\ \text{K}$).

2. Dynamic AC Resistance (r_d):
Taking the derivative dV_D/dI_D yields:

$$r_d = \frac{26\ \text{mV}}{I_D}$$

At $I_D = 2\ \text{mA}$:

$$r_d = \frac{26\ \text{mV}}{2\ \text{mA}} = 13\ \Omega$$

3. Calculate LED Series Resistor (R):
Applying KVL around the loop:

$$R = \frac{V_{CC} - V_{\text{LED}}}{I_{\text{LED}}} = \frac{12\ \text{V} - 2.0\ \text{V}}{15 \times 10^{-3}\ \text{A}} = \frac{10\ \text{V}}{0.015\ \text{A}} = 666.67\ \Omega$$

Standard practical resistor value: $680\ \Omega$.

EXAM PITFALL & CAUTION:

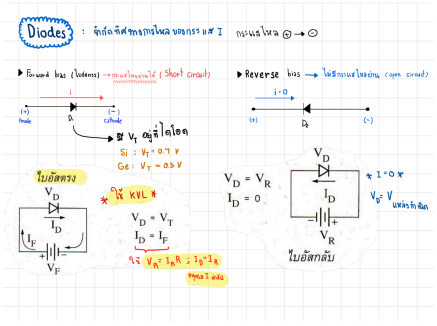
Never connect an LED directly to a voltage source without a current-limiting resistor, as diode current rises exponentially and destroys the LED.

CH10-06 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam

Zener Diode Voltage Regulator Operation

PROBLEM / QUESTION:

How does a Zener diode function as a voltage regulator? In a circuit with Vin = 15 V to 20 V, Vz = 10 V, Pz_max = 1 W, and load RL = 500 ohms, calculate the required series resistor R.



รูปประกอบ: Zener Diode Voltage Regulator Circuit

KEY FORMULAS & DEFINITIONS:

$$I_L = \frac{V_Z}{R_L} = \frac{10\ \text{V}}{500\ \Omega} = 20\ \text{mA}$$

$$I_{Z,\max} = \frac{P_{Z,\max}}{V_Z} = \frac{1\ \text{W}}{10\ \text{V}} = 100\ \text{mA}$$

$$R_{\min} = \frac{V_{\text{in,max}} - V_Z}{I_{Z,\max} + I_L} = \frac{20 - 10}{0.100 + 0.020} = 83.3\ \Omega$$

$$R_{\max} = \frac{V_{\text{in,min}} - V_Z}{I_{Z,\min} + I_L} = \frac{15 - 10}{0.005 + 0.020} = 200\ \Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Zener Regulation Mechanism:
A Zener diode operates in the REVERSE breakdown region. When reverse-biased beyond V_Z, terminal voltage remains clamped at V_Z over a wide range of reverse current I_Z.

2. Load Current (I_L):

$$I_L = \frac{V_Z}{R_L} = \frac{10\ \text{V}}{500\ \Omega} = 20\ \text{mA} = 0.020\ \text{A}$$

3. Maximum Zener Current ($I_{Z,\max}$):

$$I_{Z,\max} = \frac{P_{Z,\max}}{V_Z} = \frac{1\ \text{W}}{10\ \text{V}} = 100\ \text{mA} = 0.100\ \text{A}$$

4. Calculate Minimum Series Resistor ($R_{\min}$):
To protect the Zener from exceeding maximum power when V_{in} is highest (20 V):

$$R_{\min} = \frac{V_{\text{in,max}} - V_Z}{I_{Z,\max} + I_L} = \frac{20\ \text{V} - 10\ \text{V}}{0.100\ \text{A} + 0.020\ \text{A}} = \frac{10\ \text{V}}{0.120\ \text{A}} = 83.3\ \Omega$$

5. Calculate Maximum Series Resistor ($R_{\max}$):
To maintain regulation at lowest V_{in} (15 V) assuming $I_{Z,\min} = 5$ mA:

$$R_{\max} = \frac{V_{\text{in,min}} - V_Z}{I_{Z,\min} + I_L} = \frac{15\ \text{V} - 10\ \text{V}}{0.005\ \text{A} + 0.020\ \text{A}} = \frac{5\ \text{V}}{0.025\ \text{A}} = 200\ \Omega$$

Choose standard resistor $R = 120\ \Omega$ or $150\ \Omega$.

EXAM PITFALL & CAUTION:

Zener diode must be connected in REVERSE BIAS (Cathode to positive supply) to regulate voltage. In forward bias, it behaves like an ordinary 0.7 V diode!

CH10-07 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 9 & 10 (Renewables)

Solar Photovoltaic Cell Equivalent Circuit & Parameters

PROBLEM / QUESTION:

Draw and explain the electrical equivalent circuit of a Solar Photovoltaic Cell. Define Open-Circuit Voltage (Voc), Short-Circuit Current (Isc), Maximum Power Point (MPP), and Fill Factor (FF).

KEY FORMULAS & DEFINITIONS:

$$I = I_{\text{ph}} - I_D - I_{\text{sh}} = I_{\text{ph}} - I_s\left(e^{\frac{V+IR_s}{nV_T}} - 1\right) - \frac{V + IR_s}{R_{\text{sh}}}$$

$$P_{\max} = V_{\text{mp}} \cdot I_{\text{mp}}$$

$$\text{Fill Factor: } FF = \frac{P_{\max}}{V_{oc} \cdot I_{sc}} = \frac{V_{\text{mp}} \cdot I_{\text{mp}}}{V_{oc} \cdot I_{sc}}$$

$$\text{Efficiency: } \eta = \frac{P_{\max}}{P_{\text{in}}} = \frac{V_{oc} \cdot I_{sc} \cdot FF}{G \cdot A_{\text{cell}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Solar Cell Equivalent Circuit Components:
Consists of a light-generated photocurrent source I_{ph} in parallel with:
- An intrinsic PN junction diode D
- A shunt resistance R_{sh} (leakage path across cell edges, ideally ∞)
- A series resistance R_s (bulk silicon and contact resistance, ideally 0).

2. Key Parameters Defined:
- Short-Circuit Current (I_{sc}): Current when terminals are shorted ($V = 0$):

$$I_{sc} \approx I_{\text{ph}}$$

- Open-Circuit Voltage (V_{oc}): Voltage when net current is zero ($I = 0$), typically ≈ 0.6 V per silicon cell.
- Maximum Power Point ($P_{\max}$): Optimal operating point on the I-V curve:

$$P_{\max} = V_{\text{mp}} \cdot I_{\text{mp}}$$

- Fill Factor (FF): Ratio of peak power to theoretical product ($V_{oc}I_{sc}$):

$$FF = \frac{V_{\text{mp}}I_{\text{mp}}}{V_{oc}I_{sc}}$$

Typically ranges from 0.75 to 0.85 for high-quality commercial silicon cells.

EXAM PITFALL & CAUTION:

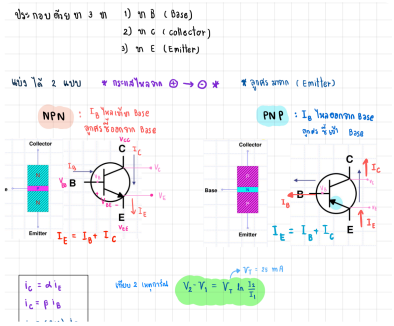
Single silicon solar cell produces only ~0.55 - 0.6 V. That is why 60 or 72 cells are connected in series inside a solar panel to output ~36 - 40 V.

CH10-08 | Chapter 10 | Level 1 (Core Concept) | Lecture Chapter 10 & Homework

BJT Transistor Terminal Currents & Gain Relations

PROBLEM / QUESTION:

State the current relationship between the Emitter (IE), Base (IB), and Collector (IC) of a Bipolar Junction Transistor (BJT). Define DC current gain Beta (hFE) and Alpha, and derive the formula linking Beta and Alpha.



รูปประกอบ: BJT Transistor Terminal Currents & Alpha-Beta Relations

KEY FORMULAS & DEFINITIONS:

$$I_E = I_B + I_C$$

$$\beta = \frac{I_C}{I_B} \qquad \alpha = \frac{I_C}{I_E}$$

$$\alpha = \frac{\beta}{\beta + 1} \qquad \beta = \frac{\alpha}{1 - \alpha}$$

STEP-BY-STEP MODEL SOLUTION:

1. Kirchhoff's Current Law for BJT:
By KCL, total current passing through the emitter equals the sum of base and collector currents:

$$I_E = I_B + I_C$$

2. DC Current Gains Defined:
- Common-Emitter Gain (β or h_{FE}):

$$\beta = \frac{I_C}{I_B} \quad (\text{typically } 50 - 300)$$

- Common-Base Gain (α):

$$\alpha = \frac{I_C}{I_E} \quad (\text{slightly less than 1, typically } 0.98 - 0.998)$$

3. Mathematical Relationship Derivation:
Dividing $I_E = I_B + I_C$ by I_C:

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1$$

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1 = \frac{\beta + 1}{\beta}$$

Inverting yields:

$$\alpha = \frac{\beta}{\beta + 1} \qquad \beta = \frac{\alpha}{1 - \alpha}$$

EXAM PITFALL & CAUTION:

IE is always the LARGEST current in a BJT: IE = IB + IC. Collector current is slightly smaller than emitter current (Alpha < 1).

CH10-09 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Lab Experiments

BJT Transistor as an Electronic Switch

PROBLEM / QUESTION:

Explain how an NPN BJT operates as an electronic switch. Define Cutoff and Saturation states, and design a base resistor RB to drive a 12 V, 100 mA relay from a 5 V microcontroller GPIO pin (assume Beta_sat = 10, Vbe_sat = 0.7 V, Vce_sat = 0.2 V).

KEY FORMULAS & DEFINITIONS:

$$\text{Cutoff (OFF): } V_{\text{in}} = 0 \implies I_B = 0,\ I_C = 0,\ V_{CE} = V_{CC}$$

$$\text{Saturation (ON): } I_B \geq \frac{I_{C,\text{sat}}}{\beta_{\text{sat}}} \implies V_{CE} = V_{CE,\text{sat}} \approx 0.2\ \text{V}$$

$$I_{C,\text{sat}} = \frac{V_{CC} - V_{CE,\text{sat}}}{R_{\text{load}}}$$

$$R_B = \frac{V_{\text{in}} - V_{BE,\text{sat}}}{I_B}$$

STEP-BY-STEP MODEL SOLUTION:

1. Operating States as a Switch:
- Cutoff (Switch OFF): When $V_{\text{in}} = 0$ V ($V_{BE} < 0.7$ V), base current $I_B = 0$ and collector current $I_C = 0$. The transistor behaves as an open switch ($V_{CE} = V_{CC}$).
- Saturation (Switch ON): Ample base current I_B forward-biases both B-E and B-C junctions. V_{CE} drops to $V_{CE,\text{sat}} \approx 0.2$ V, acting as a closed contact.

2. Base Resistor Calculation:
- Required collector load current: $I_{C,\text{sat}} = 100\ \text{mA} = 0.100\ \text{A}$.
- Required base current for deep saturation (using forced $\beta_{\text{sat}} = 10$):

$$I_B = \frac{I_{C,\text{sat}}}{\beta_{\text{sat}}} = \frac{100\ \text{mA}}{10} = 10\ \text{mA} = 0.010\ \text{A}$$

- Calculate base resistor R_B:

$$R_B = \frac{V_{\text{in}} - V_{BE,\text{sat}}}{I_B} = \frac{5.0\ \text{V} - 0.7\ \text{V}}{0.010\ \text{A}} = \frac{4.3\ \text{V}}{0.010\ \text{A}} = 430\ \Omega$$

Standard practical resistor: $430\ \Omega$ or $390\ \Omega$.

EXAM PITFALL & CAUTION:

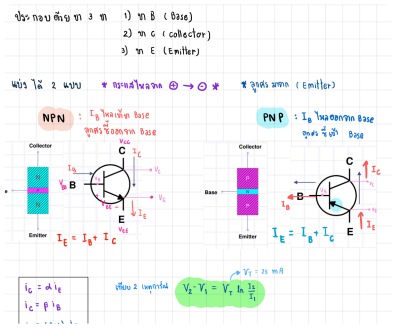
When designing a switching transistor, never use the linear data-sheet Beta (e.g. 100-300). Always use forced Beta_sat = 10 to ensure deep saturation!

CH10-10 | Chapter 10 | Level 1 (Core Concept) | Final Exam ข้อ 4.2 & Lecture Chapter 10

BJT vs FET / MOSFET Detailed Comparison

PROBLEM / QUESTION:

Provide a comprehensive comparison between Bipolar Junction Transistors (BJT) and Field-Effect Transistors (FET / MOSFET) regarding carrier conduction, control mechanism, input impedance, thermal stability, and noise.



รูปประกอบ: Transistor BJT vs MOSFET Comparison

KEY FORMULAS & DEFINITIONS:

$$\text{BJT: Bipolar (Holes + Electrons) | Current-Controlled } (I_B) | R_{\text{in}} \approx 1 - 10\ \text{k}\Omega$$

$$\text{MOSFET: Unipolar (Majority only) | Voltage-Controlled } (V_{GS}) | R_{\text{in}} \approx 10^{12} - 10^{14}\ \Omega$$

STEP-BY-STEP MODEL SOLUTION:

Comprehensive Comparison:

1. Carrier Conduction:
- BJT is BIPOLAR (current conduction involves both electrons and holes simultaneously).
- MOSFET is UNIPOLAR (current conduction relies strictly on majority carriers in the channel).

2. Control Mechanism:
- BJT is CURRENT-CONTROLLED: Output collector current I_C is controlled by input base current I_B ($I_C = \beta I_B$).
- MOSFET is VOLTAGE-CONTROLLED: Output drain current I_D is controlled by gate-to-source voltage V_{GS}.

3. Input Impedance (R_{in}):
- BJT: Moderate ($1\ \text{k}\Omega - 10\ \text{k}\Omega$) because the B-E junction is forward-biased.
- MOSFET: Near-infinite ($10^{12} - 10^{14}\ \Omega$) due to the insulating SiO_2 gate dielectric.

4. Thermal Stability:
- BJT has negative temperature coefficient for V_{BE}, making it vulnerable to Thermal Runaway.
- MOSFET has positive temperature coefficient for $R_{DS(\text{on})}$, preventing thermal runaway.

EXAM PITFALL & CAUTION:

BJT = Bipolar, Current-controlled, Low Rin. MOSFET = Unipolar, Voltage-controlled, Extremely high Rin.

CH10-11 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam

Half-Wave vs Full-Wave Rectifiers & PIV Rating

PROBLEM / QUESTION:

Compare Half-Wave, Full-Wave Center-Tapped, and Full-Wave Bridge Rectifiers regarding DC average output voltage, ripple frequency, efficiency, and Peak Inverse Voltage (PIV) rating across diodes.

KEY FORMULAS & DEFINITIONS:

$$\text{Half-Wave: } V_{\text{dc}} = \frac{V_m}{\pi} \approx 0.318\,V_m \qquad f_{\text{ripple}} = f_{\text{in}} \qquad \text{PIV} = V_m$$

$$\text{Center-Tapped: } V_{\text{dc}} = \frac{2V_m}{\pi} \approx 0.637\,V_m \qquad f_{\text{ripple}} = 2f_{\text{in}} \qquad \text{PIV} = 2V_m$$

$$\text{Bridge Rectifier: } V_{\text{dc}} = \frac{2V_m}{\pi} \approx 0.637\,V_m \qquad f_{\text{ripple}} = 2f_{\text{in}} \qquad \text{PIV} = V_m$$

STEP-BY-STEP MODEL SOLUTION:

1. Half-Wave Rectifier (1 Diode):
- DC Average Voltage: $V_{\text{dc}} = V_m/\pi = 0.318\,V_m$
- Ripple frequency: $f_{\text{ripple}} = f_{\text{in}}$ (50 Hz)
- Maximum theoretical efficiency: 40.6%
- Diode PIV: $\text{PIV} = V_m$

2. Center-Tapped Full-Wave Rectifier (2 Diodes):
- DC Average Voltage: $V_{\text{dc}} = 2V_m/\pi = 0.637\,V_m$
- Ripple frequency: $f_{\text{ripple}} = 2f_{\text{in}}$ (100 Hz, easier to filter)
- Maximum theoretical efficiency: 81.2%
- Diode PIV: $\text{PIV} = 2V_m$ (Major drawback: diodes must withstand twice peak voltage!)

3. Bridge Full-Wave Rectifier (4 Diodes):
- DC Average Voltage: $V_{\text{dc}} = 2V_m/\pi = 0.637\,V_m$
- Ripple frequency: $f_{\text{ripple}} = 2f_{\text{in}}$ (100 Hz)
- Maximum theoretical efficiency: 81.2%
- Diode PIV: $\text{PIV} = V_m$ (Dominant industry standard).

EXAM PITFALL & CAUTION:

Crucial exam question: Center-tapped full wave requires PIV = 2*Vm, whereas Bridge rectifier requires only PIV = Vm.

CH10-12 | Chapter 10 | Level 1 (Core Concept) | Lecture Chapter 10 & Lab Rules

Operational Amplifier (Op-Amp 741) Characteristics

PROBLEM / QUESTION:

State the ideal characteristics of an Operational Amplifier (Op-Amp): input impedance, output impedance, open-loop gain, bandwidth, and slew rate. Explain the concept of Virtual Short / Virtual Ground.

KEY FORMULAS & DEFINITIONS:

$$R_{\text{in}} = \infty \implies I_+ = I_- = 0$$

$$R_{\text{out}} = 0 \qquad A_{OL} = \infty \qquad \text{Bandwidth} = \infty$$

$$\text{Virtual Short: } V_+ \approx V_- \quad \text{(Under negative feedback)}$$

$$\text{Inverting Gain: } A_v = -\frac{R_f}{R_{\text{in}}} \qquad \text{Non-inverting Gain: } A_v = 1 + \frac{R_f}{R_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Ideal Op-Amp Characteristics:
- Input Impedance (R_{in}): Infinite ($I_+ = I_- = 0$).
- Output Impedance (R_{out}): Zero (supplies load without drop).
- Open-Loop Gain (A_{OL}): Infinite ($10^5 - 10^6$ in real chips).
- Bandwidth (BW): Infinite (flat response from DC to ∞).
- Slew Rate: Infinite (instantaneous output slew).

2. Virtual Short & Virtual Ground:
Because open-loop gain A_{OL} is infinite, in any closed-loop negative feedback configuration, the differential input voltage must be zero:

$$V_+ - V_- = \frac{V_{\text{out}}}{A_{OL}} = \frac{V_{\text{out}}}{\infty} = 0 \implies V_+ = V_-$$

This condition is known as a Virtual Short. If the non-inverting terminal V_+ is connected to physical ground (0 V), then V_- is held at zero volts without being physically grounded, known as Virtual Ground.

EXAM PITFALL & CAUTION:

Virtual short (V+ = V-) holds ONLY when negative feedback is present. In an open-loop comparator, V+ is NOT equal to V-.